WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS AND TESTING DIVISION

MATERIALS PROCEDURE

GUIDE FOR QUALITY CONTROL AND ACCEPTANCE PLANS FOR SUBGRADE, BASE COURSE, AND AGGREGATE ITEMS

1. PURPOSE

1.1 The purpose of this Materials Procedure (MP) is to establish minimum requirements for the Contractor's Quality Control (QC) Program and Acceptance Plan. It is intended that these requirements be used as a procedural guide in detailing the inspection, sampling, and testing deemed necessary to maintain compliance with the material and Specification requirements.

1.2 To establish procedural guidelines for approval and documentation of the Master QC Plan.

2. SCOPE

2.1 This procedure is applicable to Aggregate items placed in the field. It outlines the quality control procedures for items used and includes procedures for approving and using a Master and/or Project Specific Quality Control (QC) Plan. This procedure also aids in documentation and retention of the QC Plan in ProjectWise.

3. REFERENCED DOCUMENTS

3.1 MP 109.00.21 - Basis for Charges for Non-Submittal of Sampling & Testing Documentation by the Established Deadline

3.2 MP 300.00.51 - Procedural Guidelines for Maintaining Control charts for Aggregate Gradations

3.3 MP 700.00.54 - Procedure for Evaluating Quality Control Sample Test Results with Verification Sample Test Results

3.4 MP 700.00.06 - Aggregate Sampling Procedures

3.5 ML-25, Procedure for Monitoring the Activities Related to Sieve Analysis of Fine and Coarse Aggregate

3.6 WV Division of Highways Construction Manual, Current Edition

4. GENERAL REQUIREMENTS

4.1 The Contractor shall provide and maintain a QC system that will provide reasonable assurance that all materials and products submitted to the District for acceptance will conform to the contract requirements whether natural, manufactured or processed by the Contractor or procured from suppliers, subcontractors, or vendors. The Contractor shall perform or have performed the inspections and tests required to substantiate product conformance to contract document requirements and shall also perform or have performed all inspections and tests otherwise required by the contract. The Contractor's QC inspections and tests shall be documented and shall be available for review by the Engineer/District throughout the life of the contract. The Contractor shall maintain standard equipment and qualified personnel as required by the Specifications to assure conformance to contract requirements. Procedures will be subject to the review of the District before the work is started.

5. QUALITY CONTROL PLAN

5.1 The Contractor shall prepare a QC Plan detailing the type and frequency of inspection, sampling, and testing deemed necessary to measure and control the various properties of materials and construction governed by the Specifications. As a minimum, the sampling and testing plan shall detail sampling location, sampling techniques, and test frequency to be utilized. Attachment #1 shows guidelines for the QC Plan. QC sampling and testing performed by the Contractor may be utilized by the District for acceptance.

5.1.1 A QC Plan must be developed by the Contractor and submitted to the Engineer/District prior to the start of construction on every project. Acceptance of the QC Plan by the Engineer/District will be contingent upon its concurrence with these guidelines.

5.2 As work progresses, an addendum(s) may be required to the QC Plan to keep the QC program current. Personnel may be required to show proof of certification for testing.

5.3 QC Plan Guidelines: The QC plan shall include but not be limited to the following information:

5.3.1 Name of company official responsible for QC program. Contact phone number(s) and email(s) shall be included in the cover letter.

5.3.2 List certified personnel as specified in Section 106 of the Specifications, whether from the submitting company, consultant testing firm, or both.

5.3.3 List of the Aggregate items to be controlled by QC Plan.

5.3.4 Sampling and Testing Plan: As a minimum, the sampling and testing plan should detail sampling locations, test methods, and test frequencies to be used. To facilitate the District's monitoring activities, which are described in Section 7.1, all completed gradation samples must be retained by the Contractor until further disposition is designated by the District Materials Supervisor. The QC Plan should state where and how these samples will be maintained. Applicable sections of Materials Letter ML-25 should be used for guidance.

5.3.5 Testing Facility: The plan shall state the specific location where the samples(s) will be tested and retained.

5.3.6 Documentation Plan: The Contractor's plan to document and distribute test results shall be described.

5.3.7 Forms and Distribution: Approved processing forms available on the MCS&T Webpage[1] shall be used to record the test data. Gradation tests will be recorded on Form T300. The lab

[1] https://transportation.wv.gov/highways/mcst/Pages/tbox.aspx

oratory reference number will always start with a "C" for all QC samples taken and tested by the Contractor. One copy of each completed form should be retained by the Contractor until the work is completed and accepted. The original signed copy of the test data is to be delivered to the District Materials Supervisor. To be an effective QC function, tests must be completed and results distributed in a regular and timely manner. The plan, therefore, must state what action will be taken in the event that testing and reporting are not completed in a reasonable period of time - preferably within 72 hours after the sample is taken (at the discretion of the District).

5.3.8 Control Charts: The Specifications require the plotting of gradation test results on control charts using the moving average concept as described in MP 300.00.51. The QC Plan should state where and how the charts shall be maintained and made available to District personnel. These charts are part of the District's acceptance procedures and must be available to the District when the project is completed or at the request of the District personnel. At the Contractor's request, the requirement of Control Charts may be waived on a per project basis. The Contractor will submit a written request to the District asking that the Control Charts be waived. The District will make a determination based on the size of the project and the number of gradation tests required.

5.3.9 Disposition of Non-Specification Material: The Contractor shall provide a detailed plan of action for the immediate notification of all parties involved in the event that nonconforming situations are detected.

5.3.10 Delivery Tickets

Each truckload of aggregate delivered at the project shall be accompanied by delivery ticket with all the following information:

1. Ticket number
2. Producer/Supplier Code
3. Producer/Supplier Name
4. Producer/Supplier Location
5. Contract Identification Number (CID #)
6. Federal Project Number (If applicable)
7. State Project Number
8. Date/Time
9. Material Code/Name
10. License Number of Haul Unit or Truck Number
11. Load Number
12. Gross Truck Weight
13. Tare Truck Weight
14. Net Weight
15. Weighperson's Name certifying that all information on the ticket is correct.

The following information shall be documented on the ticket by the project:

1. Contract Item Number
2. Contract Line Number

5.3.10.1 Documentation shall be provided to the project as per the requirements of Section 109.20 of the Specification.

5.3.11 Types of QC Plans

5.3.11.1 QC Plans which are intended for use on more than one project shall be defined as Master QC Plans. Section 6.1 outlines the procedures for Master QC Plan submittal and approval.

5.3.11.2 QC Plans which are intended for use on a single project shall be defined as Project Specific QC Plans. Project Specific QC Plans shall contain a cover letter which includes the following: project description, CID#, and Federal and/or State Project Number.

5.3.11.3 A contractor may submit a project specific cover letter referencing the Master QC plan instead of a Project Specific QC Plan.

5.3.11.4 Once any QC Plan is approved for a project, the key-date shall be entered in Site Manager by the appropriate District Materials personnel. The first date entered shall be the date the Project QC Plan letter is received. The second date shall be when the District approves the QC Plan for use on the project.

6. MASTER QUALITY CONTROL PLAN

6.1 The intent of a Master QC Plan is to facilitate the approval process in a more uniform manner. The Contractor may submit a Master QC when their workload in a given District is routinely repetitive for the year. Testing includes both performing the test and submitting the results as per MP 109.00.21.

6.1.1 The Contractor may submit a new Master Aggregate Items QC Plan each year to each District in which they have or expect to have work (see Attachment #2 for an example). If the Contractor does not have work or does not have a history of work in a given District for the year, then a Master Field QC Plan shall not be submitted to that District.

6.1.2 The District will review the submitted Master QC Plan to see if it meets the requirements for the Aggregate Items in the QC Plan as per Section 5.3. If accepted, the District shall assign a laboratory reference number to the Master QC Plan for future referencing. The District will acknowledge approval of each Master QC Plan to the Contractor by letter (see Attachment #3 for an example), which will include the laboratory reference number and a copy of the approved Master QC Plan. This will then be scanned and placed in ProjectWise under the appropriate District's Org for that Contractor and/or Producer/Supplier.

6.1.3 Once a project has been awarded, if a Contractor elects to use the approved Master Aggregate Items QC Plan on that project, the Contractor shall submit a letter requesting to use the Master QC Plan for that project. This letter must be on the Contractor's letterhead, be addressed to the District Engineer/Manager or their designee, and contain the following information: project number, CID#, project description, type of QC Plan, and the laboratory reference number for the Master QC Plan. (See Attachment #4 for an example).

6.1.4 The District shall review the referenced Master QC Plan to ensure it covers all items in the project. If the referenced Master QC Plan is found to be insufficient for some items on the project, the District shall request the Contractor to submit additional information for QC of those items as an addendum on a project specific basis. When the District is satisfied with the QC Plan for this project, a letter shall be sent to the Contractor acknowledging approval (see Attachment #5 for an example), with the following attached: the Contractor's project QC Plan request letter and the Master QCP approval letter. This shall then be placed in the project's incoming-mail mailbox in ProjectWise.

6.1.5 A Master QC Plan that has been approved for project use shall be good for the duration of that project, even if that project continues into future calendar years.

6.1.6 For the use of District Personnel, the District approval letter for this project must state the ProjectWise link to the referenced Master QC Plan for that Contractor. For example, WVDOT ORGS > District Organization #> Materials > Year>Master QC Plans, etc.

6.1.7 The Master Aggregate items QC Plan shall be valid for the duration of one calendar year beginning on January 1st and ending on December 31st.

7. ACCEPTANCE PLAN

7.1 Per 307.2 of the Specifications, the acceptance (verification) sampling and testing is the responsibility of the District and QC tests are the responsibility of the Contractor. Acceptance activities (sampled and tested at the frequency given in Section 7.1.2) may be accomplished by conducting verification sampling and testing completely independent of the Contractor and, in some cases, by witnessing tests performed by the Contractor, or by a combination of the two. The following guidelines provide a system, which should result in sufficient confidence in the Contractor's documentation of their QC operations to permit acceptance of the material in accordance with the procedure set forth in the Specifications.

7.1.1 The District shall review all information supplied by the Contractor on the QC Plan. Note, in particular, the qualifications of the sampler, tester, the location, and other qualifying statements about the testing facility. In the event that little qualifying information is supplied or has been demonstrated by the testing facility: Prior to work, the District (or their representative) shall review the availability, type, and suitability of the testing equipment and verify all calibrations. This information should be documented and kept available at the District Materials Section.

7.1.2 The District shall sample and test, completely independent of the Contractor, at a frequency equal to or greater than ten (10) percent of the frequency for testing given in the approved QC Plan. Witnessing the Contractor's sampling and testing activities may also be a part of the acceptance procedure, but only to the extent that such tests are considered "in addition to" the ten (10) percent independent tests.

7.1.3 Plot the results of gradation tests performed by the District on the Contractor's QC charts with a red circle, but do not include these values in the moving average. When the Contractor's tests are witnessed, circle the Contractor's test result on the control chart with red. These values are used in the moving average calculations. The laboratory number will always start with an "M" for all acceptance (verification) samples taken and tested in this manner by the District, and will always start with a "0" for all of the Contractor's tests, which are witnessed by the District.

7.1.4 Evaluate the results of acceptance (verification) tests, whether performed or witnessed by the District, in accordance with MP 700.00.54.

7.2 If the evaluation indicates similarity with the QC test(s), the control chart will be considered acceptable to that point.

7.2.1 If dissimilarity is determined, an immediate investigation shall be conducted in an effort to determine the cause. Until the situation is resolved, any samples held in accordance with ML-25 will be retained and may be used in whatever manner deemed appropriate during the investigation.

7.3 Implement ML-25 for aggregate gradations.

8. ABSENT TESTING OF MATERIAL

8.1 If the Contractor fails to perform testing of the material in accordance with the Contractor's Division Approved Quality Control Plan, payment for the portion of the item represented by the absent test shall be withheld, pending the Engineer's decision whether or not to allow the material to remain in place. Testing includes both performing the test and submitting the results as per MP 109.00.21.

8.1.1 If the Engineer allows the material to remain in place, the Division shall not pay for the material represented by the absent test. However, the Division shall pay for the cost of the placement of the material, including labor and equipment. The invoice or material supplier cost (if applicable), determined at the time of shipment, shall be used to calculate the cost of material when evaluating the total cost of labor and equipment.

Michael Mance Digitally signed by Michael Mance Date: 2025.04.23 15:09:51 -04'00'

Michael A. Mance, P.E.
Director
Materials Control, Soils & Testing Division

MP 307.00.50 Steward – Aggregate and Soils Section
MM: R
ATTACHMENTS

ATTACHMENT #1 - GUIDELINES FOR CONTRACTOR'S QUALITY CONTROL

Item Description	Property	Minimum Frequency
207 Subgrade	Gradation	One (1) sample per day of placement. Note 1
	Atterberg Limits	From an approved aggregate source: one (1) test at the beginning of placement and then each 10,000 tons. Not from an approved aggregate source a minimum of one (1) test per 6 days placement.
212 select Material for Backfill	Gradation	Minimum of one (1) sample per day of Placement. Note 1
307 Crushed Aggregate	Gradation	One (1) sample per each one-half (1/2) day placement. Note 1
	Atterberg limits	One(1) test at the beginning of placement and then each 10,000 tons thereafter
	Other tests as requested by the Division or required by the contract documents: percent crushed particles, unit weight, etc.	As requested by the Division or required by the contract documents.
307 Aggregate Shoulder course for Resurfacing Projects	Gradation	One (1) sample per day of placement. Note 1
	Atterberg limits	One (1) test at the beginning of placement and then each 10,000 tons thereafter
	Other tests as requested by the Division or required by the contract documents: percent crushed particles, unit weight, etc.	As requested by the Division or required by the contract documents.

ATTACHMENT #1 GUIDELINES FOR CONTRACTOR'S QUALITY CONTROL (CONTINUED)

315 Trail Surface Aggregate	Gradation	Minimum of one (1) sample per half day of placement.
	Atterberg Limits	From an approved aggregate source: one (1) test at the beginning of placement and then each 10,000 tons. Not from an approved aggregate source a minimum of one (1) test per 6 days placement.
604 Class 1 Aggregate	Gradation	Minimum of one (1) sample per day of placement. Note 1
606 Aggregate for Underdrain	Gradation	Minimum of one (1) sample per day of placement. Note 1
609 Bed Course Material	Gradation	Minimum of one(1) sample per day of placement. Note 1
626 Aggregate	Gradation	Minimum of one (1) sample per day of placement. Note 1
	Atterberg Limits	From an approved aggregate source: one (1) test at the beginning of placement and then each 10,000 tons. Not from an approved aggregate source a minimum of one (1) test per 6 days placement.
636 Aggregate	Gradation	One (1) sample per each one-half (1/2) day of placement. Note 1: Note 2
	Atterberg Limits	One (1) test at beginning of placement and then each 10,000 tons thereafter. Note 2

Note 1: In the event project activities are such that relatively small quantities of material are being placed per placement date, and to prevent over sampling, the Engineer may approve the following alternate sampling method: A minimum of One (1) sample per six (6) consecutive days shall be taken to represent up to each 170 cubic yards (250 tons). Sampling is to be done on the first day of aggregate placement. In this case the sample shall be taken at a random time and place

Note 2: When Aggregate for maintaining traffic is not to be part of any succeeding base or pavement course, the appropriate aggregate size shall be determined by the Engineer. If the aggregate is from an approved source, then it shall be accepted by visual inspection. If the Contractor elects to use aggregate from an unapproved source, test results shall be provided to show that the liquid limit and plasticity index meet the requirements in Table 704.6.2B

***** ATTACHMENT #2 - EXAMPLE GUIDE FOR AGGREGATE ITEMS QUALITY CONTROL PLAN *****

The Acme Company
20 First St.
Somewhere, WV XXXXXXX

Mr/./Ms/Mrs. ____________________
WV Division of Highways
District ___ Engineer/Manager
__________, WV

RE: "year" Master Aggregate Items QC Plan
DISTRICT: _____

Dear Mr./Ms/Mrs. ________________

We are submitting our Master QC Plan for Aggregate Items , developed in accordance with the (year) WVDOH Standards and Specifications, (year)WVDOH Supplemental specifications, MP300.00.51, MP 700.00.54, ML-25, and AASHTO Testing standards.

The Quality Control Program is under the direction of ________________ . They can be contacted by telephone number ____________, email ________________ and/or in person.

1.) All testing will be performed by qualified personnel as per WVDOH Specification Section 106 Control of Materials. Proof of personnel certification shall be provided to WVDOH inspectors upon request.

2.) Specify items to be controlled and the methods by which each item will be tested (For example:207,307…etc) Attachment #1 summarizes the different materials, minimum frequencies, and the appropriate test procedure or method for controlling each material.

   - 207 Items   - 212 Items   -307 Crushed Aggregate Items   - ETC>>>>>

3.) List the location (address) and lab where testing will be performed.

4.) State the method and means by which that Contractor will document and distribute test results.

5.) State what forms will be used for tests the time frame for completing testing and distributing of test information to District Materials.

6.) Specify in the QC Plan where and how the charts will be maintained and made available to Division/District personnel. Control Charts will use the moving average concept as described in MP 300.00.51.

7.) Specify a plan of action providing for immediate notification of all parties involved in the event that nonconforming material situations are detected.

Very Truly Yours,

____
Title

********** ATTACHMENT #3 WVDOH LETTERHEAD **********

ACME Company
20 First St.
SOMEWHERE, WV #####

RE: Aggregate Items Master QC Plan
Description: (Year) Construction Season

Dear Mr./Ms/Mrs. ____________________,

Your Master Aggregate Quality Control Plan **(M#-#####)** for ________________ has been reviewed and found to be acceptable for the following items:

- 207 Aggregate Items
- 212 Aggregate Items
- 307 Aggregate Items
- ETC

As work progresses throughout the season, an addendum(s) may be required to this QCP to keep the QC program current. **Also note that personnel may be required to show proof of certification for testing. Please use Lab Reference # M#-###### when corresponding about this QC plan.** Please make sure that all appropriate personnel have a copy of this plan in their possession.

Very Truly Yours,

____
Title

********** ATTACHMENT #4 - EXAMPLE **********

THE ACME COMPANY INC.
20 First St.
Somewhere, WV XXXX

Mr./Ms/Mrs ______________________
WV Division of Highways
District ___ Engineer/Manager
__________, WV __________

Subject: Aggregate Items QC plan
For project

Fed. Project No ____________________________
State Project No. _________________________
Contract ID No. _________________________
Description ____________________________

Dear Mr./Ms/Mrs. _____________________,

We would like to use our approved Aggregate Items Master Quality Control Plan, reference number _____________ for the project referenced above. We feel that all items on the referenced project are covered by the Master Quality Control Plan for Aggregate Items.

The QC Plan is under the direction of _________________________________, _______________(title), and will be the Company's contact representative to the Division of Highways District Materials and Construction Departments. They can be contacted in person at the project, by telephone ______________________________ or at email account _________________.

Very Truly Yours,

____
Title

*************** **ATTACHMENT #5 - WVDOH LETTERHEAD** **************

THE ACME COMPANY INC.
20 First St.
Somewhere, WV XXXXX

RE: _____ Aggregate Items QC Plan

Project CID#: ########
Fed/State Project #: ####- ## - ####.##
Description: Falling Slide
County : XXXXXXX

Dear Mr./Ms/Mrs. ____________________,

Your request to use your Master Aggregate Items Quality Control Plan **(M# - ######)** for Aggregate Items on the project referenced above, has been reviewed and found to be acceptable for the following items:

- 207 Aggregate Items	- 212 Aggregate Items
- 307 Aggregate Items	- ETC

As work progresses throughout this project an addendum(s) may be required to this QCP to keep the QC program current. **Please use M# - ###### when corresponding about this QC plan. Also note that personnel may be required to show proof of certification for testing.** Please make sure that all appropriate personnel have a copy of this plan in their possession.

For Division/District use

The Master Quality Control Plan can be reviewed in ProjectWise at this Link:

WVDOT ORG>D0#>year>MASTER QC PLANS>Contractors or Plant>Contractor Name>Name of Quality Control Plan

Very Truly Yours,

____
Title

WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS AND TESTING DIVISION

MATERIALS PROCEDURE

GUIDE TO DESIGNING HOT-MIX ASPHALT USING THE MARSHALL DESIGN METHOD

1. PURPOSE

1.1 To establish an approved Marshall design method, test procedures, and evaluation criteria for hot-mix asphalt (HMA). If reclaimed asphalt pavement (RAP) is used in the design, refer to Materials Procedure (MP) 401.02.24 for additional guidelines.

2. SCOPE

2.1 This procedure is applicable to design tests conducted for the purpose of establishing mixture proportions for HMA using the Marshall mix design method. Marshall designs that have already been approved under the previous version of this MP may still be used as long as the mix design verification and quality control requirements of MP 401.02.27 can be met using Tables 1, 2, and 3 of this MP as the reference design criteria. Note that Table 1 has slightly modified the air void design criteria for Base-I so verification and quality control for older designs will be based on the new value.

2.2 Any approved mix design that exhibits poor field performance may be rejected from further use by the Division.

3. REFERENCED DOCUMENTS

3.1 AASHTO Standards:

a) R 30, Mixture Conditioning of Hot Mix Asphalt (HMA)
b) T 30, Mechanical Analysis of Extracted Aggregate
c) T 164, Quantitative Extraction of Asphalt Binder from Hot Mix Asphalt (HMA)
d) T 166, Bulk Specific Gravity of Compacted Hot Mix Asphalt (HMA) Using Saturated Surface-Dry Specimens
e) T 209, Theoretical Maximum Specific Gravity and Density of Hot Mix Asphalt (HMA)T 245, Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus
f) T 269, Percent Air Voids in Compacted Dense and Open Asphalt Mixtures
g) T 308, Determining the Asphalt Binder Content of Hot Mix Asphalt (HMA) by the Ignition Method

3.2 ASTM Standards:

a) D 5581, Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus (6 inch-Diameter Specimen)

3.3 Asphalt Institute:

a) MS-2 Manual, Mix Design Methods for Asphalt Concrete and Other Hot-Mix Types – This well written Asphalt Institute reference guide explains the entire Marshall Method design process in a logical order. The mix designer must still adhere to WVDOH design property requirements and procedures, and they must use the latest AASHTO and ASTM test methods.

3.4 Material Procedures:

a) MP 401.02.24, Guide to Designing Hot Mix Asphalt with Reclaimed Asphalt Pavement.
b) MP 401.02.27, Guide for Contractor Quality Control of Hot Mix Asphalt
c) MP 700.00.06, Aggregate Sampling Procedures
d) MP 700.00.54, Procedure for Evaluating Quality Control Sample Test Results with Verification Sample Test Results

4. TESTING REQUIREMENTS

4.1 The laboratory performing the design shall be a Division approved laboratory. To obtain Division approval, a laboratory must demonstrate that they are equipped, staffed and managed, for batching and testing HMA in accordance with this MP. This shall be accomplished by submitting a copy of their latest report of inspection by the AASHTO Materials Reference Laboratory (AMRL) to the District Materials Section. The laboratory must also submit a letter detailing the actions taken to correct any deficiencies noted in the test procedures listed below. The District will forward this information to Materials Control, Soils and Testing Division (MCS&T). It is also required that the laboratory request to be included on AMRL's routine schedule of inspections, which is usually every 18 to 24 months in order to maintain their approval status.

4.1.1 AASHTO Test Procedures

a) T 245, Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus
b) T 166, Bulk Specific Gravity of Compacted Hot Mix Asphalt (HMA) Using Saturated Surface-Dry Specimens
c) T 209, Theoretical Maximum Specific Gravity and Density of Hot Mix Asphalt (HMA)
d) T 27, Sieve Analysis of Fine and Coarse Aggregates (Note 1)
e) T 11, Materials Finer Than 75 µm (No. 200) Sieve in Mineral Aggregates by Washing (Note 1)
f) T 84, Specific Gravity and Absorption of Fine Aggregate
g) T 85, Specific Gravity and Absorption of Coarse Aggregate

Note 1: T 30, Mechanical Analysis of Extracted Aggregate, may be substituted for T 27 and T 11 if the laboratory is using T 308, Determining the Asphalt Binder Content of Hot Mix Asphalt (HMA) by the Ignition Method or T 164, Quantitative Extraction of Asphalt Binder from Hot Mix Asphalt (HMA).

4.2 The laboratory is required to have a technician who has attended and successfully completed a Division approved Marshall mix design class. In addition to the class that is offered through the West Virginia University Asphalt Technology Program, hands-on Marshall mix design classes offered by the Asphalt Institute, National Center for Asphalt Technology (NCAT), National Asphalt Pavement Association (NAPA), Chicago Testing Laboratory, and various state DOTs have been approved. Proof of successful completion of all class requirements (including a written examination) must be provided. Approval of an older design class that did not require a written examination will be on a case-by-case basis including a review of the designer's experience. MCS&T will maintain a list of the approved design laboratories and design technicians.

4.3 The required mix design properties are:

4.3.1 Stability and Flow: AASHTO T 245 or ASTM D 5581 as applicable.

4.3.2 Air Voids: AASHTO T 269

4.3.3 Voids in Mineral Aggregate (VMA): Asphalt Institute MS-2 Manual

4.3.4 Voids Filled with Asphalt (VFA): Asphalt Institute MS-2 Manual

4.3.5 Fines to asphalt (FA) ratio: Asphalt Institute MS-2 Manual

4.4 The design PG Binder shall normally be selected in accordance with Section 401.2 of the Standard Specifications. However, the laboratory's mix designer should refer to the contract documents to determine if a nonstandard binder has been specified for the project.

4.5 A series of test specimens shall be prepared for a range of different asphalt contents so that the test data curves show a well-defined "optimum" value. Samples shall be fabricated to include a range of asphalt contents of at least 2 percent at intervals not to exceed 0.5 percent.

4.6 Test specimens shall be fabricated from materials of the same sources and types as proposed in the job mix formula (JMF). The gradation of the combined aggregates used in the test samples shall be the same as that proposed in the plant mix formula and shall meet the requirements of Table 401.4.2A of the Standard Specifications. The percent passing each sieve contained in Table 401.4.2A, from one sieve larger than the nominal maximum size down to the 75 µm (No. 200) sieve, shall be included in all gradation calculations.

4.7 The gradation of each aggregate size from each source used in the mix design shall be determined from an average of at least three individual gradations of each material from the stockpile at the plant or from material supplied by the aggregate producer. The aggregates shall be sampled in accordance with MP 700.00.06.

4.8 If a mix contains reclaimed asphalt pavement (RAP), the asphalt must be removed from the RAP for gradation analysis by the ignition oven method (T 308) or a solvent extraction process (T 164). If the T 164 solvent extraction test method is used, a non-chlorinated solvent may be substituted for the standard specified solvent, and the test method may be modified as per the recommendations of the solvent supplier. The solvent must be a product that has been tested for use in extracting asphalt from HMA. The RAP used for designing a mix must come from the plant stockpile from which it will be produced.

4.9 A minimum of three compacted test specimens for each combination of aggregates and asphalt content are required.

4.10 The maximum specific gravity shall be based on the average of two samples prepared at the estimated optimum asphalt content.

4.11 Immediately after mixing each of the Marshall bulk specific gravity samples and the maximum specific gravity samples, age the samples for 2 hours ± 5 minutes in accordance with AASHTO R30 before further testing.

4.12 Mixtures shall be designed in accordance with the criteria set forth in Table 1, 2, and 3 unless otherwise indicated in a special provision or as a note in the contract documents.

TABLE 1—Marshall Method Mix Design Criteria

Design Criteria	Medium Traffic Design (Note 2 and 3)	Heavy Traffic Design	Base-I Design (Note 4)
Compaction, number of blows each end of specimen	50	75	112
Stability (Newtons) (minimum)	5,300	8,000	13,300
Flow (0.25 mm) (Note 5)	8 to16	8 to 14	12 to 21
Percent Air Voids	4.0	4.0	4.0
Percent Voids Filled with Asphalt (Note 6)	65 to 80	65 to 78	64 to 73
Fines-to-Asphalt Ratio	0.6 to 1.2		

Note 2: If the traffic type is not provided in the contract documents, contact the District to obtain this information before developing the mix designs.

Note 3: All Wearing-III mixes shall be designed as a 50-blow mix regardless of traffic type.

Note 4: All Base-I mixes will be designed and tested using 112 blows with six-inch diameter specimens in accordance with ASTM D 5581.

Note 5: When using a recording chart to determine the flow value, the flow is normally read at the point of maximum stability just before it begins to decrease. This approach works fine when the stability plot is a reasonably smooth rounded curve. Some mixes comprised of very angular aggregates may exhibit aggregate interlocking which causes the plot to produce a flat line at the peak stability before it begins to drop. This type of plot is often difficult to interpret, and sometimes the stability will even start increasing again after the initial flat line peak.

When such a stability plot occurs, the stability and flow value shall be read at the initial point of peak stability.

Note 6: A Wearing-I heavy traffic design shall have a VFA range of 73–78 percent. A Wearing-III mix shall have a VFA range of 75–81 percent.

TABLE 2—Percent Voids in Mineral Aggregate (Note 7)

Mix Type	Nominal Size Sieve	Percent Voids in Mineral Aggregate (VMA) (minimum)
Wearing-III & Scratch-III	4.75 mm (No. 4)	17
Wearing-I & Scratch-I	9.5 mm (⅜ in.)	15
Base-II, P&L & Wearing-IV	19 mm (¾ in.)	13
Base-I	37.5 mm (1 ½ in.)	11

Note 7: Mixtures designed with the VMA exceeding the minimum value by more than two percent may be susceptible to flushing and rutting problems, especially when used on pavements subjected to slow moving traffic conditions. They may also be difficult to compact as they often have a tendency to shove under the roller.

TABLE 3—Design Aggregate Gradation Requirements for Marshall Mixtures (Note 8)

TYPE OF MIX	**Base-I**	**Base-II (Patch & Level)**	**Wearing-IV** (Note 9)	**Wearing-I (Scratch-I)**	**Wearing-III (Scratch-III)**
SIEVE SIZE	**Nominal Maximum Size**				
	1 ½ in (37.5 mm)	**¾ in** (19 mm)	**¾ in** (19 mm)	**3/8 in** (9.5 mm)	**No. 4** (4.75 mm)
2 in (50 mm)	100				
1 ½ in (37.5mm)	90 - 100				
1 in (25 mm)	90 max	100	100		
¾ in (19 mm)	-	90 – 100	90 – 100		
½ in (12.5 mm)	-	90 max	90 max	100	
3/8 in (9.5 mm)	-	-	-	85 - 100	100
No. 4 (4.75 mm)	-	-	47 min	80 max	90 - 100
No. 8 (2.36 mm)	15 – 36	20 – 50	20 – 50	30 – 55	90 max
No. 16 (1.18 mm)	-	-	-	-	40 - 65
No. 30 (600 µm)	-	-	-	-	-
No. 50 (300 µm)	-	-	-	-	-
No. 200 (75 µm)	1.0 – 6.0	2.0 – 8.0	2.0 – 8.0	2.0 – 9.0	3.0 – 11.0

Note 8: For quality control of the mixture the allowable tolerances for each JMF shall be the specified design control points shown in Table-3 with the exception that a Wearing-III mix shall have a tolerance limit of the JMF ± 5% on the 1.18 mm (No. 16) sieve, and all other mix types shall have a tolerance limit of the JMF ± 6% on the 2.36 mm (No.8) sieve. These tolerances shall also be applied to the mix design and shall be documented on the T-400 Form. The tolerances shall not fall outside of the specified control points of Table-3.

Note 9: In addition, a Wearing-IV mix shall have a tolerance limit of the JMF ± 5% on the 4.75 mm (No. 4) sieve, but not below the minimum requirement.

5. DETERMINING THE OPTIMUM ASPHALT CONTENT

5.1 Prepare a graphical plot of the following relationships:

a) Asphalt Content vs. Percent Air Voids.
b) Asphalt Content vs. Stability
c) Asphalt Content vs. Flow
d) Asphalt Content vs. VMA
e) Asphalt Content vs. VFA

5.2 From the plot of asphalt content vs. percent air voids, pick the asphalt content that corresponds to the 4.0 percent air voids.

5.3 If the corresponding stability, flow, VMA, and VFA values are within the specified design criteria at the asphalt content determined in Section 5.2, then this asphalt content shall be considered the optimum asphalt content for the mix.

5.4 If the design property values determined as per Section 5.3 do not meet the specified criteria at the percent asphalt content determined in Section 5.2, then new mix proportions must be determined, and new test data developed.

5.5 Full mix design testing will not be required when a mix design is developed using the sources, exact aggregate types, and compaction level as a prior Division approved design, along with a different neat binder grade. The designer may instead select to make a set of bulk specific gravity test specimens and a maximum specific gravity test specimen with the approved aggregate structure and the new binder grade at the optimum asphalt content of the approved design. Since these samples are laboratory produced design specimens, they must be oven aged for 2 hours ± 5 minutes before testing in accordance with R 30. Mix and compaction temperature will be based on the requirements of the new binder grade. The percent air voids must be 4.0 ± 0.3 percent. The voids-in-mineral aggregate must be within ± 0.5 percent of the original approved job mix formula design target (but not outside of the design limits of this MP). All other mix design criteria must be within the design limits specified in this MP (including stability and flow). If the mix design meets all of these requirements then this test data may be submitted along with a new T400 form for approval as a new mix design. A copy of the approved T400 on which this new design is based should also be included. If the mix design fails to meet all of the requirements then a new mix design must be developed.

6. REPORT

6.1 The T-400 JMF form shall include the design property information required in Section 401.4 of the Standard Specifications. The JMF package shall include the following:

6.1.1 A summary sheet (Marshall Mix Design Package Attachment #1, Optimum Asphalt Content Determination) showing the proposed asphalt content determination plus the design properties compared to the design criteria of Table 1. This attachment shall be signed and dated by the mix design technician.

6.1.2 The chart showing the plots described in Section 5.1 used to determine the optimum asphalt content (Attachment #2).

6.1.3 A Summary of Marshall Mix Design Data worksheet (Attachment #3).

6.1.4 Worksheet for calculating the effective gravity of the blended aggregates (Attachment #4 or #4A).

6.1.5 Worksheets showing calculations for maximum specific gravities of the mix at different asphalt contents (Attachment #5). For any mix design that contains any single coarse aggregate component with the water absorption of 1.5 percent or greater, follow the supplemental procedure of T 209 to determine if a dry-back is necessary. Because the dry-back procedure is addressing an aggregate coating issue, this same supplemental procedure shall be used on quality control and verification samples of mixes containing these high absorptive aggregates to determine if the dry-back procedure is necessary.

6.1.6 Worksheet for calculating the bulk and apparent specific gravities of the total aggregate, and the percent VMA in the compacted mixture (Attachment #6 or #6A).

6.1.7 Worksheet for determining the maximum specific gravity of the mixture, including the dry-back procedure when required (Attachment #7).

6.1.8 Worksheets showing calculation for bulk and apparent specific gravities and absorption of the coarse and fine aggregates used in the mix design (Attachments #8 and #8A).

6.1.9 The 0.45 power gradation chart (Attachment #9) developed for each mix design. This chart shall include the maximum density line, aggregate control points, and a gradation plot showing each screen used in the design.

6.1.10 A worksheet showing the calculations for the combined aggregate of the mix design (Attachment #10).

6.1.11 Worksheets showing the washed sieve analysis results for each aggregate used in the mix design (Attachment #11).

6.1.12 The temperature-viscosity chart for the asphalt used in the mix design. An asphalt supplier issued chart or document containing the mix and compaction temperature recommended for the specific grade of asphalt will be acceptable.

6.2 The entire printed JMF package shall be submitted to the local District Materials Section in which the HMA plant is located. After reviewing, the District shall attach a memo to the JMF package requesting approval of the design and submit it to the MCS&T Asphalt Section.

6.2.1 The JMF package can also be submitted electronically by scanning it into an Adobe Acrobat Reader file and e-mailing the file to the appropriate District Materials Section and the MCS&T Asphalt Section. After reviewing the JMF package, the District will send an e-mail to the MCS&T Asphalt Section verifying that the JMF package has been reviewed. The District will also note any problems that they find with the JMF. The MCS&T Asphalt Section will conduct a final review on the design package and assign a laboratory number to each approved mix design. MCS&T will contact the mix designer if there are any problems or concerns with the JMF package that will delay final approval. An electronic copy of the approved T400 form shall be e-mailed to the District and Producer for distribution.

6.3 All applicable mix design worksheets can be found on the MCS&T's Webpage[1] under the "Toolbox."

01/04/2023

Ronald L. Stanevich, PE
Director
Materials Control, Soils & Testing Division

MP 401.02.22 Steward – Asphalt Section
RLS:J

[1] https://transportation.wv.gov/highways/mcst/Pages/default.aspx

WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS AND TESTING DIVISION

MATERIALS PROCEDURE

GUIDE TO DESIGNING ASPHALT MIXTURES WITH RECLAIMED ASPHALT PAVEMENT

1. PURPOSE

1.1 To establish criteria for designing asphalt mixtures which contain reclaimed asphalt pavement (RAP) and Performance Graded (PG) Binders.

2. SCOPE

2.1 This procedure is applicable to all asphalt mixture designs which contain both RAP and PG Binders.

3. GENERAL

3.1 This MP does not alter the design specification requirements of the 401 Specification or MP 401.02.22. It is to be used only as a supplement to the specifications when designing RAP mixtures. It does not affect RAP mixtures which were designed through previously approved methods prior to issuance of this MP.

4. REFERENCED DOCUMENTS

4.1 MP 401.02.22

5. GUIDELINES

5.1 The following guidelines shall apply to all new mix designs which incorporate RAP with PG Binders.

5.2 For design purposes, the specific gravity of the virgin PG Binder shall be used as the specific gravity of the asphalt binder in the RAP. Also, the effective specific gravity of the aggregate in the RAP shall be determined and used as the bulk specific gravity of the RAP aggregate for calculation purposes.

5.3 If the amount of RAP in the mix is equal to or less than 15 percent, then the selected PG Binder to be used as the virgin asphalt shall be the same as the specified PG Binder for the region where the mix will be used. For example, if the specified PG Binder for the region is a PG 64S-22 then the PG Binder used in the RAP design shall be a PG 64S-22.

5.4 If the amount of RAP in the mix is 16 to 25 percent, then the selected PG Binder to be used as the virgin asphalt shall be one grade below both the high and low temperature grade of the specified PG Binder for the region where the mix will be used. For example, if the specified PG Binder for the region is a PG 64S-22 then the PG Binder used in the RAP design shall be a PG 58S-28.

5.5 If the amount of RAP in the mix is more than 25 percent, then the blending chart described in Section 6.0 of this MP shall be used to select the high temperature grade of the virgin asphalt. The low temperature grade shall be at least one grade lower than the binder grade specified for the area where the mix will be used. The binder test data and the blending chart must be submitted along with the mix design package (JMF).

6. EXAMPLE USE OF BLENDING CHART

6.1 The dynamic shear rheometer (DSR) can be used to look at permanent deformation (rutting factor) of the binder, which is governed by limiting G*/sin d at the test temperature. The maximum allowable value of the rutting factor shall be 2.0 kPa. A blending chart, similar to the viscosity blending charts used with viscosity graded asphalts, has been developed which plots G*/sin d on a log-log scale on the y-axis as a replacement test for viscosity. Both the recovered asphalt and the virgin asphalt are tested at the high temperature of the specified binder to be used in the design. The test value G*/sin d for each asphalt is plotted on the chart (the recovered asphalt result on the left and the virgin asphalt on the right) and connected with a straight line. The point on the chart where the plot of G*/sin d intersects the y-axis (G*/sin d, kPa, at test temperature) at 2.0 kPa is represented on the x-axis (% virgin binder) as the minimum percentage of virgin binder to be used in the RAP design.

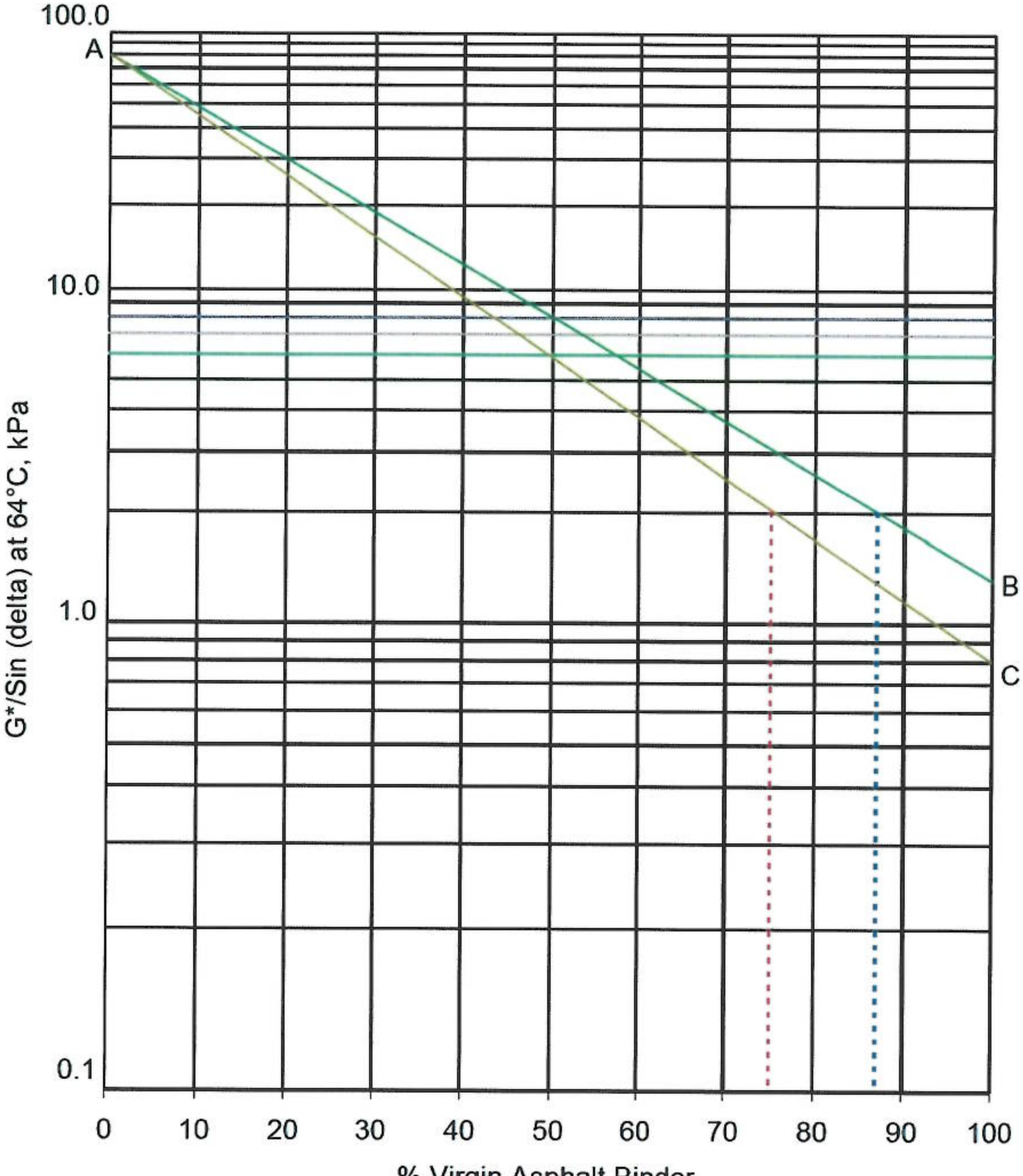
6.2 The attached example illustrates how the blending chart shall be used. The standard binder for the design in this example is a PG 64S-22. Test measurements for both the recovered asphalt and the virgin binder are taken at 64 °C. Point A on the chart represents the G*/sin d value for the recovered asphalt. Point B represents the G*/sin d test value for the PG 64S-22 binder which has a minimum requirement of 1.0 kPa. The line connecting points A and B intersects the 2.0 kPa rutting factor value at approximately 87% on the x-axis. This means that the minimum amount of virgin asphalt (PG 64S-22) used in the RAP design shall be 87%.

6.3 Looking at point C on the example chart, this represents G*/sin d for a PG 58S-28 Binder which has been tested at 64 °C. A PG 58S-28 Binder would normally be tested at 58 °C and would have a minimum G*/sin d value of 1.0 kPa. However, because this material is being used where the required binder is a PG 64S-22, the virgin binder must be tested at 64 °C. The higher test temperature results in a test value of less than 1.0 kPa, as illustrated on the chart attachment. The line connecting points A and C intersects the 2.0 kPa rutting factor value at approximately 75% on the x-axis. This means that the minimum amount of virgin asphalt (PG 58S-28) used in the RAP design shall be 75%.

01/04/2023
Ronald L. Stanevich, P.E.
Director
Materials Control, Soils and Testing Division

MP 401.02.24 Steward – Asphalt Section
RLS:J
ATTACHMENT

PG Binder/RAP Blending Chart



WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS AND TESTING DIVISION
MATERIALS PROCEDURE

CERTIFICATION OF ASPHALT SHIPPING TERMINALS

1. PURPOSE

1.1 To establish procedures and conditions for certifying asphalt shipping terminals and to establish inspection and shipping procedures for certified terminals.

2. SCOPE

2.1 This procedure applies to liquid asphalt products furnished to state projects and purchase orders for use in highway construction and maintenance. However, Materials Control, Soils and Testing (MCS&T) may elect to use other control procedures when special conditions dictate the need for more stringent control.

3. GENERAL

3.1 This standard may involve hazardous materials, operations and equipment. It does not address all of the safety problems associated with their use. The user of this standard will be responsible for appropriate safety and health practices.

3.2 The materials covered by this Material Procedure and the applicable specification requirements are as follows:

Material	WVDOH Specification Section	Applicable AASHTO Specification
Performance Graded Asphalt Binders	705.5	AASHTO M-332
Asphalt Emulsions (Anionic)	705.4	AASHTO M 140 and M 316
Asphalt Emulsions (Cationic)	705.11 and 705.12	AASHTO M 208 and M 316
Asphalt Plastic Cement	708.9	None

4. TERMINOLOGY

4.1 AASHTO - The American Association of State Highway and Transportation Officials.

4.2 AASHTO:resource - The AASHTO Materials Reference Laboratory formerly AMRL.

4.3 COA – Certificate of Analysis.

4.4 HMA – Hot Mix Asphalt.

4.5 Lot – A batch of material produced.

4.6 MCS&T – Materials Control, Soils and Testing.

4.7 MP – Material Procedure.

4.8 WVDOH - West Virginia Department of Highways

5. SIGNIFICANCE AND USE

5.1 This Material Procedure (MP) sets forth a method for the quality control of liquid asphalt products. This is accomplished by a certification system that uses test data from both the supplier and the Division.

5.2 This MP provides information on the following activities:

5.2.1 General requirements that the supplier must satisfy for approval under the certification program.

5.2.2 Minimum requirements to be included in the Supplier's quality control plan.

5.2.3 Procedures for shipping asphalt.

5.2.4 Procedure for MCS&T monitoring of the system.

5.2.5 Procedure for evaluating monitor, supplier, and field sample test data.

6. PROCEDURE

6.1 QUALITY CONTROL PLAN - The supplier will submit a quality control plan, to ensure that the material in stock complies with AASHTO:resource specification requirements. The plan will include the following:

6.1.1 Facility type (refinery, terminal, in-line blending, or HMA plant, etc.), location, person responsible for quality control at the facility.

6.1.2 The tests to be conducted, the name and location of the laboratory or laboratories conducting the tests and the frequency at which the tests are to be conducted. The supplier will include test data that will be sent to Materials Control, Soil and Testing (MCS&T), and how often it is to be submitted. Suggested frequency is weekly, or monthly, during the time when the terminal is in operation.

6.2 LABORATORY - The supplier will provide a laboratory with the necessary test equipment and personnel to test the asphalt for specification compliance. The testing may be done at more than one location, for example: terminal laboratory and refinery laboratory and the supplier may elect to have all or part of the tests conducted by a commercial laboratory.

6.2.1 The principal laboratory must be accredited by the AASHTO:resource, or have applied for such inspection and shall participate in the AASHTO:resource Proficiency Sample Testing Program. Exceptions will be made for those materials for which AASHTO:resource does not provide accreditation or a proficiency sample testing program. In such cases MCS&T may conduct its own inspection as a substitute for the AASHTO:resource inspection.

7. SAMPLING AND TESTING

7.1 Sampling and testing shall consist of Quality Control tests by the supplier and testing of monitor samples and field samples by MCS&T. MCS&T field samples, when available, will also be used in the evaluation.

7.2 The supplier will obtain samples of all materials to be shipped to West Virginia Department of Highway (WVDOH) projects, at a minimum sampling frequency of one sample each time the tank is filled, or new material is added. All tests required by the specifications will be conducted on each batch.

7.3 The supplier will evaluate the test results. If any lot of asphalt does not meet specifications, it must be reworked or reblended until it does meet specifications, or that lot (batch) of material must not be shipped to WVDOH projects. The test data will be submitted to MCS&T.

7.4 MCS&T will obtain and test monitor samples, of all materials to be shipped to WVDOH projects. An MCS&T inspector or consultant will visit each supplier approximately three times per year for the purpose of collecting liquid asphalt samples. The inspector or consultant will obtain samples of each material for which the supplier is certified, or wishes to become certified, and has in stock. Additional visits to obtain samples may be made if additional data is needed to evaluate a specific material. These samples will be tested for compliance with all requirements of the governing specifications. Suppliers must include a Certificate of Analysis (COA) for all materials certified at the time of sampling.

7.5 MCS&T will evaluate the Quality Control data, monitor sample data and any field sample data that may have been obtained, at intervals of approximately 30 days. If the asphalt does not meet the required criteria, a review will be made of the entire system. This may include sampling and testing procedures, and resampling and testing of both Quality Control and MCS&T samples. If this does not resolve the problem and the asphalt still does not meet the criteria, future shipments of that asphalt must be made using the procedure given for Level Two, in Section 8.3 of this MP, until the criteria is met.

7.5.1 If the material continues to not meet specifications or if the supplier has not manufactured a certified material or materials for a maximum of 2 years, then MCS&T retains the right to remove the certification from that material(s).

7.6 Performance Graded Binders, Asphalt Emulsions and Asphalt Plastic Cement shall adhere to the WVDOH Specifications 705 and/or AASHTO:resource.

7.7 FIELD SAMPLES - Field samples will be tested for compliance with the governing specifications. If the material does not meet these criteria, it will not be shipped under Level Two procedures until the criteria is met.

7.8 The percent within tolerance for field samples is defined as the percent of the material statistically predicted to be within the specification limits. It may be based on either the normal distribution, or the t-distribution, as applicable for the sample size. The minimum number of samples for statistical evaluation is four. For fewer samples the evaluation of failing samples will be made on an individual basis.

7.9 When the requirements for certification have been met, the Division will notify the supplier. Shipments may then be made using the procedure given for Level One Quality Procedure, in Section 8.1 of this MP.

8. SHIPPING PROCEDURES

8.1 LEVEL ONE - To qualify for Level One shipment, the material must be from a certified terminal, or refinery, and must be included on the current list of approved Asphalt Materials. Shipments may be made at any time.

8.2 The supplier will prepare a shipping invoice containing the following information: name and location of company, type and grade of asphalt, quantity and date shipped, and a statement that the asphalt meets specifications. In addition, for material shipped by tanker, the invoice will contain a statement that the transport vehicle has been inspected for contamination and has been found to be acceptable for the type of material being shipped.

8.3 LEVEL TWO - Level Two shipments will consist of shipments of asphalt that are not included in the current list of Certified Asphalt Materials, or shipments that are made from unapproved terminals.

8.4 Each lot will be sampled by MCS&T or its consultant. The quantity represented by the sample will be the quantity in the storage tank at the time the sample is taken. A new sample must be taken when new material is added to the tank. In the case of materials stored in drums, or pails, the quantity represented by the sample will be the quantity of that lot on hand when the sample was taken.

8.5 If the sample meets specifications, shipments may be made until the entire lot has been shipped, or in the case of materials stored in tanks, until new material is added to the tank.

8.6 Shipments may be made in accordance with paragraph 8.2 of this MP, except that the specific lot of material must have been tested and must meet specifications. The following additional information is required with the invoice: lot or tank number, date shipped and destination.

8.7 Asphalt that does not meet specifications may not be shipped until it has been reworked or reblended and meets specifications.

10/28/2021

Ronald L. Stanevich, P.E.
Director
Materials Control, Soils and Testing Division

RLS: Jp

WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS AND TESTING DIVISION

MATERIALS PROCEDURE

GUIDE FOR CONTRACTOR QUALITY CONTROL OF ASPHALT CONCRETE

1. PURPOSE

1.1 To provide a method for daily monitoring and quality control of Asphalt Concrete.

1.2 To provide plant personnel with criteria upon which to base decisions of continuing or ceasing plant production.

2. SCOPE

2.1 This materials procedure shall be applicable to all Section 401 Asphalt Concrete types relative to compliance with Job Mix Formula (JMF) control limits as specified in the governing specifications.

3. DEFINITIONS

3.1 Job Mix Formula - The specification for a single mix produced at a single plant. This mix may be specific to a single project or be used on multiple projects if the basic design criteria (design compaction level and PG Binder grade) are the same.

3.2 Field Design Verification Samples and Tests - Those samples taken and tests conducted by the contractor to verify that a mix design can be produced within the limits of the criteria set forth by this Materials Procedure. These samples are taken during the initial use of each mix design or whenever circumstances described in this MP require a new field design reverification. These samples should not be confused with the Division verification samples that are used to determine specification compliance.

3.3 Quality Control Samples and Tests - Those samples taken and tests conducted by the Producer/Contractor to monitor and control the production of this product.

3.4 Verification Samples and Tests - Those samples taken and tests conducted by the Division to determine specification compliance.

4. DOCUMENTATION

4.1 The Contractor shall maintain adequate records of all testing and records of any production changes required to control their product. The records shall indicate the nature and number of observations made, the number and type of deficiencies found, and the nature of corrective action taken. The Contractor's documentation procedures will be subject to the review and approval of the Division at any time during the progress of the work being performed.

4.2 Forms and Distribution: All test data shall be documented on forms provided by the Division. The original copy of the completed form shall be delivered to the District Materials Supervisor. One copy of each completed form is to be retained by the Contractor until the project is completed. Testing shall be conducted using only the approved test methods listed in Section 401.5.1 of the Standard Specification unless specified otherwise in contract documents. Asphalt content and gradation test results shall be recorded on T417. Mix design property test results shall be recorded on form T406. To be an effective quality control program, tests must be completed in a regular and timely manner.

4.3 The Contractor shall take prompt action to correct conditions that have resulted, or could result, in the submission to the Division of materials and products that do not conform to the requirements of the Contract documents. The Contractor shall establish a detailed plan of action regarding the disposition of non-specification material. In the event that non-specification material is incorporated into the project, the Division shall be notified immediately.

4.4 All Asphalt Concrete component materials shipped to the plant must have proper documentation, which identifies the type and source of each material. This information shall be made accessible to the Division for review at any time.

5. JOB MIX FORMULA FIELD DESIGN VERIFICATION

5.1 For each JMF, a field design verification shall be conducted during the first days of plant production for the purpose of demonstrating that the mix can be produced within the specified tolerances set forth in this MP.

5.2 This field design verification shall consist of a randomly selected Asphalt Concrete sample taken in accordance with AASHTO T168 for each three hours of production, with no more than three samples in one day. A minimum of three samples are required for verification, however, three additional samples are required if none of the first three samples are completely within the specification limits. Samples used for gradation analysis during the verification process shall be obtained from the asphalt ignition oven samples (AASHTO T308). If there is a problem with major aggregate breakdown affecting the gradation test results when using the ignition oven, gradation samples may be obtained from hot bins, cold feeds, or extracted Asphalt Concrete samples.

5.3 Field design verification testing shall not be conducted if less than 200 tons (180 Mg) of material is to be produced in a single day. In such cases daily quality control testing shall be conducted in accordance with Section 6. and shall meet the gradation requirements of the Table 401.02.27B, the design asphalt content within ± 0.4%, and a minimum VMA of 0.5% below the design criteria specified in MP 401.02.22. The percent air voids shall be within the range of 2.5 – 6.5 percent for Base-I and 2.5 – 5.5 percent for all other mixes. Stability and flow shall be within the design limits specified in MP 401.02.22.

5.4 The field design verification mix property requirements are listed in Table 401.02.27A. Field design verification test results shall be documented on Form T408. Gradation requirements for the field design verification samples shall be as indicated in Table

401.02.27B. The gradation results shall fall within the limits of each specified control point with the exceptions as noted on the No. 8 and No. 16 sieves. Gradation results for all sieves listed in this table for each mix type shall be documentedon Form T421.

TABLE 401.02.27A

Mix Property Field Design Verification Requirements

Property	Field Verification Tolerances
Asphalt Content (%)	JMF ± 0.4 %
Air Voids (%) – Base-I	3.0 – 6.0 %
Air Voids (%) – All other mix types	3.0 – 5.0 %
Voids in Mineral Aggregate (VMA) %	Min. of 0.5 % Below Design Criteria
Stability (Newtons)	Minimum Design Criteria
Flow (0.25 mm)	Limits of Design Criteria

TABLE 401.02.27B

Design Aggregate Gradation Requirements for Marshall Mixtures (Note 8)

TYPE OF MIX	Base-I	Base-II (Patch & Level)	Wearing-IV (Note 9)	Wearing-I (Scratch)	Wearing-III
SIEVE SIZE	Nominal Max Size 1 ½ in (37.5 mm)	Nominal Max Size ¾ in (19 mm)	Nominal Max Size ¾ in (19 mm)	Nominal Max Size 3/8 in (9.5 mm)	Nominal Max Size No. 4 (4.75 mm)
2 in (50 mm)	100				
1 ½ in (37.5 mm)	90 – 100				
1 in (25 mm)	90 max	100	100		
¾ in (19 mm)	-	90 – 100	90 – 100		
½ in (12.5 mm)	-	90 max	90 max	100	
3/8 in (9.5 mm)	-	-	-	85 - 100	100
No. 4 (4.75 mm)	-	-	47min	80 max	90 – 100
No. 8 (2.36 mm)	15 – 36	20 – 50	20 – 50	30 – 55	90 max
No. 16 (1.18 mm)	-	-	-	-	40 – 65
No. 30 (600 µm)	-	-	-	-	-
No. 50 (300 µm)	-	-	-	-	-
No. 200 (75 µm)	1.0 – 6.0	2.0 – 8.0	2.0 – 8.0	2.0 – 9.0	3.0 – 11.0

Note 8: For quality control of the mixture the allowable tolerances for each JMF shall be the specified design control points shown in Table-3 of MP 401.02.22 with the exception that a Wearing-III mix shall have a tolerance limit of the JMF ± 5% on the 1.18 mm (No. 16) sieve, and all other mix types shall have a tolerance limit of the JMF ± 6% on the 2.36 mm (No.8) sieve. These tolerances shall also be applied to the mix design and shall be documented on the T-400 Form. The tolerances shall not fall outside of the specified control points of Table-3 of MP 401.02.22.
Note 9: In addition, a Wearing-IV mix shall have a tolerance limit of the JMF ± 5% on the 4.75 mm (No. 4) sieve, but not below the minimum requirement.

5.5 After each of the field design verification samples is tested, the results shall be evaluated to determine conformance to the verification requirements. If any test results fall outside the allowable tolerance limits established in Table 401.02.27A or Table 401.02.27B then steps must be taken to make any necessary production adjustments to bring the mix back to within specification limits. Steps can include bin changes as described in 5.8, as well as asphalt content adjustments of ± 0.2% from the approved JMF target. If, after three samples, all of the design criteria and gradation requirements are within the allowable tolerance limits on at least one sample, then verification of the design is complete. If all criteria is not met, then three additional samples shall be tested. If, after six samples, the Division determines that the mix cannot be produced within specification limits, then a new mix design will be required.

5.6 The verified JMF target for asphalt content shall be selected at a value within ± 0.2% of the approved design asphalt content using the results of the field verification testing to determine the appropriate value. The VMA production target shall be determined from the field verification test data at a value which also provided an air void content that was at or near the JMF target air void content based on the results of the field verification testing. This value may be adjusted to optimize the ± 1.0 tolerance of Table 401.02.27C if the result is near the minimum allowable requirement. The production target for air voids shall remain at the medium value of the design.

5.7 If the field design verification process is successful, then a new target maximum density shall be established for compaction control by averaging the maximum density results of all of the samples used for verification of the mix. The District will forward the verification test data to the Contract Administration Division, Materials Section.

5.8 The maximum allowable blend change for a mix design shall be ten percent on any single aggregate component. If an aggregate blend change of more than five percent on any single aggregate component is required, the Contractor shall evaluate the mix to determine whether or not the volumetric properties, FA ratio, and coarse aggregate angularity are adversely affected by the change in blended aggregates. The Contractor shall also determine whether or not the aggregate gradation requirements are still being maintained. The calculations used in this evaluation shall be provided to the District. The District will review and verify the results of this evaluation. If the District determines that any of the above-mentioned properties are adversely affected by the blend adjustment, then they may revoke the change in the JMF. If the JMF volumetric properties cannot be maintained without these non-approved changes, then the contractor will be required to provide a new mix design.

5.9 After the field design verification has been successfully completed and quality control testing (as described in Section 6.) has begun, the Contractor shall monitor the maximum specific gravity of the mix for any consistent change. If, over a five-sample period, there is an average change in the maximum specific gravity of $\pm$ 0.02 or greater from the verified value of the mix then a field design reverification may be required. A reverification shall not be conducted if the averages of the % asphalt, % air voids, %VMA, stability and flow of the five quality control samples do not meet the requirements of Table 401.02.27C. The District will review the Contractor's test data, compare it to their verification sample test data, and determine if a reverification is necessary. If the District determines that a reverification of the mix is needed, a new blended aggregate bulk specific gravity shall also be determined for the mix before the field reverification begins. The District will forward the reverification and bulk aggregate specific gravity test results to the Contract Administration Division, Materials Section.

5.10 All approved mix designs shall be reverified on the first project on which they are used in any subsequent years as long as there are no changes to the design specifications that would require a new mix design. In addition, the blended aggregate bulk specific gravity shall be determined before reverification begins.

6. QUALITY CONTROL REQUIREMENTS

6.1 After the field design verification has been successfully completed, quality control sampling and testing shall begin. If production is to continue for four hours or more after the last field design verification sample was taken, then the first randomly selected quality control sample shall be taken within that remaining time period. If production continues for less than four hours after the last field design verification sample was taken, then the first randomly selected quality control sample will not be required until the next production day.

6.2 The allowable design property tolerances for each JMF shall be as set forth in Table 401.02.27C. The gradation of the mix shall continue to pass through the control points within the tolerances established in Table 401.02.27B.

6.3 Adjustments to the accepted JMF aggregate proportions shall be made only for the purpose of maintaining the gradation requirements of Table 401.02.27B and/or the design properties of Table 401.02.27C. The maximum allowable adjustment shall be as indicated in Section 5.8. The minimum sample requirements of the approved quality control plan will be sufficient when the allowable adjustments are made as a result of deficient or borderline test properties of the previous test sample.

TABLE 401.02.27C

Quality Control Mix Property Tolerances

Property	Production Tolerances
Asphalt Content (%)	Verified JMF ± 0.4 %
Air Voids (%)	JMF ± 1.5 %
Voids in Mineral Aggregate (VMA) %	Verified JMF ± 1.0 % with a minimum of 0.5 % below the minimum design criteria
Stability (Newtons)	Minimum Design Criteria
Flow (0.25 mm)	Limits of Design Criteria

6.4 If the previous test sample meets all specification requirements, but the Contractor later determines that the gradation of the material entering the plant has changed, then an aggregate proportion adjustment up to two percent will be allowed without requiring an additional test sample. However, if more than one such change is made during the production day, then an additional test sample beyond that specified in the approved quality control plan will be required for each adjustment.

6.5 Minimum Sampling and Testing Frequency: During each day of plant production a minimum of one sample shall be taken for production periods of six hours or less. When the production period exceeds six hours, a minimum of one sample for each half of the production period shall be taken. If the production period exceeds twelve hours, a third sample shall be taken. The Contractor's sampling frequency shall be in accordance with their approved Quality Control Plan.

6.6 For the purpose of administration, the quantity of material represented by an individual test shall be determined as follows: the first sample taken after the field design verification has been approved shall represent the quantity produced from the beginning of production after field design verification until the time the sample was taken. The second sample shall represent the material produced between the time that the first and second samples were taken and so on. The last sample taken prior to a halt in production under a given JMF shall represent that quantity of material produced from the time that the next to last sample was taken until production was stopped.

6.7 Sampling and testing for evaluation of compliance with the verified JMF shall be as follows: Obtain a sample large enough for determining the percent asphalt, percent air voids, percent VMA, and gradation of the mix in accordance with the specified test methods listed in Section 401.5.1 of the Specifications. If excessive aggregate breakdown in the ignition oven prevents proper gradation analysis, aggregate samples may be obtained from hot bins, cold feeds, or extracted Asphalt Concrete samples.

6.8 A four-sample average shall be used for the purpose of determining whether or not the material meets specification requirements. The test results of the first four samples shall be averaged. After the fifth sample is taken a four-sample moving average shall begin. This first moving average shall consist of the average of the second through fifth test samples. Each time a new sample is taken a new moving average shall be calculated by averaging the new sample with the previous three samples. The moving average shall continue through a single paving season (one calendar year).

6.9 In cases where production is limited and less than four samples of the specified mix design are taken, then the average shall consist of the total number of samples taken during the paving season in accordance with the Quality Control Plan. A new four sample average shall be established at the first startup of a new paving season after the field design verification has been completed.

6.10 The Contractor shall maintain control charts for percent asphalt, percent air voids, and percent VMA. These control charts shall be prepared in accordance with the guidelines of MP 300.00.51. As an alternative method, the control charts may be prepared with a personal computer using software that can generate such charts and provide a distinct graphic representation of all data points. Data points required on the control charts are the daily individual Contractor quality control tests, district verification sample tests, and the moving average of every four Contractor quality control tests. All data points shall be calculated to the nearest 0.1 percent.

6.11 For hand drawn charts, the quality control test data points shall be represented by a small blue circle symbol "O" and connected by a dashed line. The four sample moving average data points shall be represented by a small red square symbol "■" and connected by a solid line. District verification sample test data points shall be represented by a small red circle symbol "O", but shall not be connected. The upper and lower tolerance limits of the test properties that were established through the field design verification described in Section 6. shall be represented by solid horizontal lines.

6.12 If the computer-generated control chart cannot be produced using the symbols and lines described above, then a graph legend shall be included which shall indicate the graphic symbols used to represent the required data points and lines.

6.13 The quality control charts shall be kept up to date and placed in a location that is easily accessible to the Division for review at any time.

7. DEGREE OF NONCONFORMANCE

7.1 Should the four-sample average of test values for percent asphalt, percent air voids, or percent VMA fall outside the verified JMF tolerances by more than the allowable deviation of Table 401.02.27C then production shall be halted until the Contractor takes necessary steps to bring production under control. Production shall also be halted if three consecutive aggregate gradation tests fall outside the tolerance limits of Table 401.02.27B. Actions taken by the Contractor to bring production back in control shall be documented in the plant diary. Once production starts again, the moving average is reset.

7.2 When the four sample average of the Contractor's quality control tests for percent asphalt or percent air voids falls outside the JMF tolerances of Table 401.02.27C, the Sublot of material represented by the last individual test value in the moving average shall have its price reduced in accordance with the schedule set forth in Section 7.3. In the case where the average is nonconforming and the last tested Sublot is conforming, then there would be no price adjustment.

7.3 The degree of nonconformance shall be determined using the following relationship:

When the moving average is greater than the upper control limit

$Q_U = X_n - UL$

When the moving average is less than the lower control limit

$Q_L = LL - X_n$

Where Q_U = Percent of non-conformance at Upper Limit
Q_L = Percent of non-conformance at Lower Limit
UL = Upper Limit
LL = Lower Limit
X_n = Average of four consecutive test values (less than four when production is limited)

If it is decided by the Division that the material is to be allowed to remain in place, then the Sublot shall have its price reduced in accordance with Tables 401.02.27D and/or 401.02.27E as applicable.

TABLE 401.02.27D
ADJUSTMENT OF CONTRACT PRICE FOR MIX NOT WITHIN TOLERANCE LIMITS OF PERCENT ASPHALT

QU or QL	Percent of Contract Price to be Paid
0.0	100
0.1	98
0.2	96
0.3	92
Greater Than 0.3	*

* The Division will make a special evaluation of the material and determine the appropriate action.

TABLE 401.02.27E
ADJUSTMENT OF CONTRACT PRICE FOR MIX NOT WITHIN TOLERANCE LIMITS OF PERCENT AIRVOIDS

QU or QL	Percent of Contract Price to be Paid
0.0	100
0.1	98
0.2	96
0.3	92
Greater Than 0.3	*

* The Division will make a special evaluation of the material and determinethe appropriate action.

7.4 Should the moving average of both the test properties for the same Sublot fall outside of the JMF tolerance, thus resulting in a reduced price for each, then the following procedure shall be used. The quantity of material represented by the last Sublot in the moving average will have an adjusted unit price which is the product of the original price times the percent as a result of non-conformance of the first test property times the percentage unit price as a result of non-conformance of the second test expressed in the following formula.

AUP = OUP x PUPAC x PUPAV *

Where: AUP = Adjusted Unit Price
OUP = Original Unit Price
PUPAC = Percent Unit Price as a result of Asphalt Content Analysis expressed as a decimal
PUPAV = Percent Unit Price as a result of Air Void Analysis expressed as a decimal

* PUPAC and PUPAV are used in the formula as needed as a single non-conforming item or together for both non-conforming items as shown.

7.5 A new moving average shall start with the fourth sample that is taken after production is resumed (less than four when production is limited). If, at any time, the Division determines that a mix cannot be consistently produced within the tolerance limits of the verified design properties, approval of the mix may be revoked and the contractor will be required to provide a new mix design.

8. SMALL QUANTITY TESTING

8.1 In the event that project activities are such that not more than 75 tons (70 Mg) of a specific mix design is being produced per day during the period of an entire calendar week, then the following small quantity testing requirements shall apply.

8.2 If the plant source rating is A-1, as determined per MP 700.00.52, Guide To Source Rating System Relative To Maintenance Contracts, then the minimum quality control sample requirements shall be one sample per week. The sample shall be taken on the first day of use during the week. If the plant source rating is A-2, as determined per MP 700.00.52, then the normal testing requirements of this MP shall apply.

9. DIVISION VERIFICATION SAMPLING AND TESTING

9.1 Verification sampling and testing is the responsibility of the Division. Quality control tests conducted by the Contractor may be used as a part of the verification process. Verification activities may be accomplished in any of three ways: 1) By conducting sampling and testing completely independent of the Quality Control activities, 2) by witnessing tests performed by the Contractor, or 3) by a combination of both the above. In all cases, those samples and tests taken by the Division completely independent of the Contractor will be taken at a frequency approximately equal to 10% of the frequency required in the Contractor's approved Quality Control Plan for the applicable item.

9.2 The verification samples taken by the Division will be statistically evaluated for similarity to the Contractors quality control tests in accordance with the guidelines set forth in MP 700.00.54. If the evaluation indicates that the Division's test results are similar to the Contractor's test results, then the material represented by this evaluation will be considered acceptable. Those properties to be evaluated, as referenced in MP 700.00.54, will consist of percent asphalt, percent air voids, stability, flow, and gradations. In addition, the VMA test results will be evaluated using the guidelines of MP 700.00.54.

9.3 If a dissimilarity is detected, an immediate investigation will be conducted to determine the cause. The intent of the investigation is to define and correct any testing deficiencies that may cause a misrepresentation of the tested material.

DocuSigned by:
7CC4F23D4D94414...

Ron L. Stanevich, P.E.
Director
Materials Control, Soils and Testing Division

RLS:Cb

WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS AND TESTING DIVISION

MATERIALS PROCEDURE

GUIDE FOR QUALITY CONTROL PLANS FOR ASPHALT

1. PURPOSE

1.1 This procedure presents uniform Quality Control (QC) guidelines for Contractor (and/or Producer(s) to develop their QC Plan. All items listed are believed necessary to assure adequate product QC.

1.2 This procedure also creates a more uniform process for District Materials to review and approve Quality Control Plans for use on projects.

2. SCOPE

2.1 This Material Procedure (MP) is applicable to, but not limited to the following Asphalt Items:

1. Base
2. Wearing
3. Patching and Leveling Courses
4. All P.W.L. Items
5. Skid

3. REFERENCED DOCUMENTS

3.1 MP 109.00.21 - Basis for Charges for Non-Submittal of Sampling & Testing Documentation by the Established Deadline.

4. GENERAL REQUIREMENTS

4.1 As stated in 401.6.1 of the Specifications, a QC Plan must be developed by the Contractor and/or Producer and submitted to the Engineer prior to construction. Acceptance of the Quality Control Plan by the Engineer will be contingent upon its concurrence with these guidelines. For this reason, the plan should clearly describe the methods by which the Quality Control Program will be conducted. For example, the items to be controlled, tests to be performed, testing frequencies, sampling locations and techniques all should be included and each item should be listed separately. Also, a detailed plan of action regarding disposition of non-specification material should be included. Such a plan should provide for immediate notification of all parties involved in the event non-conforming situations are detected. Attachment #1 may be used as an example Quality Control Plan for plant operations using all items that are applicable to the specific type of plant items produced.

Attachment #2 may be used as an example Quality Control Plan for field operations using all items that are applicable to field work.

4.2 Inspection and testing records shall be maintained, kept current, and made available for review by the Engineer throughout the life of the contract. All other documentation, such as date of inspections, tests performed, temperature measurements, and any accuracy, calibration, or re-calibration checks performed on production or testing equipment should be recorded.

4.3 The Contractor shall maintain standard calibrated equipment and certified personnel in accordance with contract and Specification requirements for the item(s) being produced.

4.4 The Division reserves the right to review all pertinent documents concerning equipment calibration used for testing and proof of certified personnel performing tests.

5. MASTER QUALITY CONTROL PLAN

5.1 The intent of a Master QC Plan is to facilitate the approval process in a more uniform manner. The Contractor may submit a Master QC when their workload in a given District is routinely repetitive for the year.

5.1.1 The Contractor may submit a new Master Asphalt Items QC Plan each year to each District in which they have or expect to have work. If the Contractor does not have work or does not have a history of work in a given District for the year, then a Master QC Plan shall not be submitted to that District.

5.1.2 The District will review the submitted Master QC Plans to see if they meet the requirements for the Asphalt Items in the QC Plan. If accepted, the District shall assign a laboratory reference number to the Master QC Plans for future referencing. The District will acknowledge approval of each Master QC Plan to the Contractor by letter (see Attachment #3 for an example), which will include the laboratory reference number and a copy of the approved Master QC Plan. This will then be scanned and placed in ProjectWise under the appropriate District's Org for that Contractor and/or Producer/Supplier.

5.1.3 Once a project has been awarded, if a Contractor elects to use the approved Master Asphalt Items QC Plan on that project, the Contractor shall submit a letter requesting to use the Master QC Plan for that project. This letter must be on the Contractor's letterhead, be addressed to the District Engineer/Manager or their designee, and contain the following information: project number, CID#, project description, type of QC Plan, and the laboratory reference number for the Master QC Plan. (See Attachment #4a and 4b for Plant and Field operations respectively for examples.)

5.1.4 The District shall review the referenced Master QC Plan to ensure it covers all items in the project. If the referenced Master QC Plan is found to be insufficient for some items on the project, the District shall request the Contractor to submit additional information for QC of those items as an addendum on a project specific basis. When the District is satisfied with the QC Plan for this project, a letter shall be sent to the Contractor acknowledging approval (see Attachment #5 for an example), with the following attached: the Contractor's project QC Plan request letter and the Master QCP approval letter. This shall then be placed in the project's incoming-mail mailbox in ProjectWise.

5.1.5 A Master QC Plan that has been approved for project use shall be good for the duration of that project, even if that project continues into future calendar years.

5.1.6 For the use of District Personnel, the District approval letter for this project must state the ProjectWise link to the referenced Master QC Plan for that Contractor. For example, WVDOT ORGS > District Organization #> Materials > Year>Master QC Plans, etc.

5.1.7 The Master Asphalt Items QC Plan shall be valid for the duration of one calendar year beginning on January 1st and ending on December 31st.

5.2 Delivery Tickets

5.2.1 Each truckload of asphalt delivered to the project shall be accompanied by one delivery ticket with the following items listed on the ticket:

1. Ticket Number
2. Producer/Supplier Code
3. Producer/Supplier Name
4. Producer/Supplier Location
5. Contract Identification Number (CID #)
6. Federal Project Number (If applicable)
7. State Project Number
8. Date/Load Out Time
9. Material Code/Name
10. License Number of Haul Unit or Truck Number
11. Approved Job Mix Formula (JMF) Number
12. Gross Weight (TN)
13. Tare Weight (TN
14. Net Weight (TN)
15. Daily Cumulative Weight (TN)
16. Maximum Density*
17. Lab Number for Testing**
18. Weighperson's Name certifying that all information on the ticket is correct.

*In addition, once the design has been verified, the newly established Maximum Density shall be reported on each ticket thereafter.
** This may be added by the project or District.

The following information shall be documented on the ticket by the project:

1. Contract Item Number
2. Contract Line Number
3. Truck Temperature at Jobsite (°F/°C)

5.2.1.1 Documentation shall be provided to the project as per the requirements of Section 109.20 of the Specifications.

6. ASPHALT FOR MAINTENANCE

6.1 The provisions of this MP will also apply to asphalt plant run purchase orders that are picked up at the plant by the Division's Maintenance forces. Yearly Master Plant and Field QCP's apply to Laydown Asphalt Purchase Orders awarded to vendors. Exceptions to this are as specified in the Purchase Order Maintenance Contract.

7. ACCEPTANCE PLAN

7.1 The Asphalt Material shall be accepted in accordance with the material's specific MP and the Standard Specifications.

7.2 Key Dates

7.2.1 Once the Quality Control Plan is approved for the project the key date shall be entered into the current AASHTOWare software by the appropriate District Materials personnel. The first date entered shall be the date the Project Quality Control Plan letter is received. The second date shall be when the district approves the quality control plan for use on the project.

8. ABSENT TESTING OF MATERIAL

8.1 If the Contractor fails to perform testing of the material in accordance with the Contractor's Division Approved Quality Control Plan, payment for the portion of the item represented by the absent test shall be withheld, pending the Engineer's decision whether or not to allow the material to remain in place. Testing includes both performing the test and submitting the results as per MP 109.00.21.

8.1.1 If the Engineer allows the material to remain in place, the Division shall not pay for the material represented by the absent test. However, the Division shall pay for the cost of the placement of the material, including labor and equipment. The invoice or material supplier cost (if applicable), determined at the time of shipment, shall be used to calculate the cost of material when evaluating the total cost of labor and equipment.

9. MATERIAL TEST DATA

9.1 The Contractor's Quality Control Plan shall clearly state the name(s) of the individual(s) entering test data as outlined in MP 109.00.21.

Michael Mance
Digitally signed by Michael Mance
Date: 2024.08.01 14:19:36 -04'00'

Michael A. Mance, P.E.
Interim Director
Materials Control, Soils & Testing Division

MP 401.03.50 Steward – Asphalt Section
MM: J
ATTACHMENTS

ATTACHMENT #1 – EXAMPLE QC PLAN FOR PLANT OPERATIONS

Mr./Ms/Mrs ________________________
West Virginia Division of Highways
District: ____Engineer/Manager
_______________,West Virginia

Dear Mr/Ms./Mrs ______________________

Subject: Asphalt Concrete Yearly
Master Quality Control
Plan for Plant Operations

We are submitting our asphalt concrete hot-mix asphalt Quality Control Plan, developed in accordance with Section 401 of the _________________ Standard Specifications, the ________________ Supplemental Specifications, MP 401.03.50 and the _______________Special Provisions.

1. Make of Plant Type Location

 ____

2. The Quality Control program is under the direction of____________ to be contacted at Plant/Office in person, by telephone # and /or email address:________________

3. Sampling and testing will be performed by qualified personnel as per WVDOH Specification Section 106 Control of Materials.

4. The types of asphalt paving materials to be produced and controlled are as follows:
    a. _____________
    b. _____________
    c. _____________
    d. _____________
    e. _____________
    f. _____________

5. Prior to production of the items, we will submit (on Division Form T-400) our plant mix formula for each type of mix. Only approved materials will be incorporated in the mix.

6. During the production operations of the asphalt concrete we will perform at a minimum Quality Control tests in accordance with the attached schedule.

7. All testing and evaluation will be completed within 24 hours of sampling and all documentation will be completed and submitted to the Division on approved processing forms within 72 hours or production will be halted until these items are current.

ATTACHMENT #1 – EXAMPLE QC PLAN FOR PLANT OPERATIONS (CONTINUED)

8. Material found to be noncomplying shall not be incorporated into the roadway. In the event that non-specification material is incorporated into the project, the Division of Highways District Materials Supervisor will be notified immediately.

9. We will notify all appropriate Division of Highways personnel at least 24 hours before the scheduled work is to begin.

10. (Statement of disposition of nonconforming material)

____

____

____

____

Very truly yours,

____

Company Representative

ATTACHMENT #1 – EXAMPLE QC PLAN FOR PLANT OPERATIONS (CONTINUED)

GUIDE FOR QUALITY CONTROL PLANS FOR ASPHALT CONCRETE

TEST OR ACTION	FREQUENCY	TEST METHOD	METHOD OF DOCUMENTATION
Construction of stockpile to prevent segregation intermingling	Constant	Visual	Diary
Coarse aggregate unit weight	One test before start of operation	AASHTO T19	T304
Stockpile & cold bin gradations	Plant setup and as needed to control production	AASHTO T27 and T11	T300
Calculate % aggregate from each bin, calibration cold bin	Plant setup		T415
Check feeder gate output at gate setting to be used	Plant setup. If bins overflow or run dry.		Plant Inspection Form and Diary
Select screen sizes	Plant setup		Plant Inspection Form
Determine hot bin gradation, calculate combined gradations	Weekly during production	AASHTO T27 and T11	T300 and T415
Calibrate hot bins, select gate openings, and calculate batch weights	Plant setup and change in material source		Plant Inspection Form and Diary
Check accuracy of scales	Plant setup and weekly accuracy checks. Zero balance and sensitivity each ½ day of operation	Construction Manual – 700 Series	Plant diary and T603
Calibrate asphalt pump, calculate settings	Plant setup		Plant Inspection Form
Check metering pump at setting to be used	Plant setup and monthly		Plant Inspection Form and Diary
Reset metering pump to compensate for temperature change	Plant setup and each temperature change of 10 °F (6 °C)		Plant Inspection Form and Diary
Adequate heated storage for liquid asphalt	Plant setup		Plant Inspection Form

ATTACHMENT #1 – EXAMPLE QC PLAN FOR PLANT OPERATIONS (CONTINUED)

GUIDE FOR QUALITY CONTROL PLANS FOR MARSHALL DESIGNED ASPHALT CONCRETE

TEST OR ACTION	FREQUENCY	TEST METHOD	METHOD OF DOCUMENTAT
Calculating mixing time	Plant setup and when paddle pitch or dam gate changed		Plant Inspection Form and Diary
Ross Count (degree of coating)	Only if mixing time is less than 45 seconds	AASHTO T195	Diary
Coarse aggregate face fracture (Gravel only)	One test before start of operation Every 10,000 ton (9,000 Mg) thereafter	MP 703.00.21	T302
Complete mix face fracture (When using gravel)	One per week	MP 703.00.21	T302
Check moisture content of aggregate	Plant setup and daily		Diary
Temperature check	Minimum of one check of mix per hour at plant		Plant Control Chart and Diary
Asphalt Content		AASHTO T308 (Method A)	T417, T423 and Control Charts
Aggregate Gradation (cold feed, hot bins, or completed mix)	In accordance with WVDOH MP 401.02.27	AASHTO T27 plus T11 or AASHTO T30	T300, T404, or T417 Plus T425
Daily Mix Property Testing: Stability and Flow, % Air Voids, and % Voids-in-Mineral Aggregate (VMA)		AASHTO T245 or ASTM D5581, AASHTO T269, T166, T209, and MS-2 Manual	T406 and T423

ATTACHMENT #1 – EXAMPLE QC PLAN FOR PLANT OPERATIONS (CONTINUED)

GUIDE FOR QUALITY CONTROL PLANS FOR SUPERPAVE DESIGNED ASPHALT CONCRETE

TEST OR ACTION	FREQUENCY	TEST METHOD	METHOD OF DOCUMENTATIO
Calculating mixing time	Plant setup and when paddle pitch or dam gate changed		Plant Inspection Form and Diary
Ross Count (degree of coating)	Only if mixing time is less than 45 seconds	AASHTO T195	Diary
Coarse aggregate face fracture (Gravel only)	One test before start of operation Every 10,000 ton (9,000 Mg) thereafter	ASTM D5821	T302
Complete mix face fracture (When using gravel)	One per week	ASTM D5821	T302
Check moisture content of aggregate	Plant setup and daily		Diary
Temperature check	Minimum of one check of mix per hour at plant		Plant Control Chart and Diary
Gyratory Compaction		AASHTO TP4	T419
Aggregate Gradation		AASHTO T30	T417 and T425
Asphalt Content		AASHTO T308 (Method A)	T417 and Control Charts
Percent Air Voids		AASHTO T166, T209, and T269	
Percent Voids in Mineral Aggregate (VMA)	In accordance with WVDOH MP 401.02.29	AASHTO PP-28	T419 and Control Charts
Percent Voids Filled With Asphalt (VFA)		AASHTO PP-28	

ATTACHMENT #2 – EXAMPLE QC PLAN FOR FIELD OPERATIONS

Mr./Ms/Mrs ______________________
West Virginia Division of Highways
District: ____Engineer/Manager
______________,West Virginia

Subject: Asphalt Concrete Yearly
Master Quality Control
Plan for Field Operations

Dear Mr./Ms/Mrs. __________________

We are submitting our asphalt concrete hot-mix asphalt Quality Control Plan, developed in accordance with Section 401 of the _______________ Standard Specifications, the _______________ Supplemental Specifications, MP 401.03.50 and the ______________ Special Provisions.

1. The field operation is under the direction of ___________ by telephone # and /or email address:_______________
2. ________________________________ will be responsible for insuring that all items of work will comply with Division specifications.
3. During the placement operation of the asphalt concrete pavement we will perform, at a minimum, Quality Control tests as per the attached schedule. Sampling and testing will be performed by qualified personnel as per WVDOH Specification Section 106 Control of Materials.
4. All sampling and testing will be completed within the time limits specified by the Division or work will be halted.
5. Material found to be non-complying shall not be incorporated into the roadway. In the event that non-specification material is incorporated into the project, the Division representative will be notified immediately.
6. We will notify all appropriate Division personnel at least 24 hours before work is scheduled to begin.

Very truly yours,

____
Company Representative

ATTACHMENT #2 – EXAMPLE QC PLAN FOR FIELD OPERATIONS (CONTINUED)

GUIDE FOR QUALITY CONTROL PLANS FOR FIELD

TEST OR ACTION	FREQUENCY	TEST METHOD	METHOD OF DOCUMENTATIO
Temperature of mix	1 per hour	Section 401 of Standard Specifications	Diary
Temperature of base	1 per hour	Section 401 of Standard Specifications	Diary
Temperature of mat	1 test per hour of placement	Section 401 of Standard Specifications	T401
Density	5 tests per 1000 feet (300 meters) of paving width or rollerpass when applicable.	Section 401 of Standard Specifications	T401 or T407
Tack/Prime Application Rate	Each load or per ½ day of operation whichever occurs first	Section 408/409 of Standard Specifications	Diary
Calibration of Nuclear Gauge	As per MP 717.04.21	As per MP 717.04.21	Factory Data Sheet
Distribution of Test Data	Within 24 hours of completion of testing of a Lot	As per MP 717.04.21	As per MP 717.04.21

ATTACHMENT #3 – MASTER PLAN ACCEPTANCE FOR CONSTRUCTION SEASON

ACME Company
20 First St.
SOMEWHERE, WV #####

RE: Asphalt Items Master QC Plan
Description: (Year) Construction Season

Dear Mr./Ms/Mrs. ____________________,

Your Master Asphalt Quality Control Plan **(M#-#####)** for _________________ has been reviewed and found to be acceptable for the following items:

> As work progresses throughout the season, an addendum(s) may be required to this QCP to keep the QC program current. **Also note that personnel may be required to show proof of certification for testing. Please use Lab Reference # M#-###### when corresponding about this QC plan**. Please make sure that all appropriate personnel have a copy of this plan in their possession.

Very Truly Yours,

____
Title

ATTACHMENT #4A – REQUEST TO USE MASTER QC FOR PLANT

Mr/./Ms/Mrs. ______________________
WV Division of Highways
District ___ Engineer
__________, WV _________

RE: Asphalt Concrete Quality Control plan for Plant - Project

Fed. Project No. __________________________

State Project No. __________________________

Contract ID No. __________________________

County : __________________________

Description __________________________

Dear Mr./Ms/Mrs. ____________________,

We would like to use our approved Master Quality Control Plan, reference number _____________ for the project referenced above.

The Quality Control Plan is under the direction of ________________________________, _______________(title), and will be the company's contact representative to the Division of Highways District Materials and Construction Departments. They can be contacted in person at the plant, by telephone ______________________________ or at email account ________________

Very Truly yours,

____
Company Representative

ATTACHMENT #4B – REQUEST TO USE MASTER QC FOR FIELD

EXAMPLE

Mr./Ms/Mrs ______________________
WV Division of Highways
District ___ Engineer
__________, WV ___________

RE: Asphalt Concrete Quality Control plan for Field - Project

Fed. Project No ____________________________

State Project No. _________________________

Contract ID No. _________________________

County : ____________________________

Description ____________________________

Dear Mr./Ms/Mrs. ____________________,

We would like to use our approved Master Quality Control Plan, reference number ______________ for the project referenced above.

The Quality Control Plan is under the direction of __________________________________, _______________(title), and will be the company's contact representative to the Division of Highways district materials and construction departments. They can be contacted in person at the project, by telephone ______________________________ or at email account _________________.

Very Truly yours,

____
Company Representative

ATTACHMENT #5 – DISTRICT ACCEPTANCE OF MASTER QC PLAN FOR SPECIFIC PROJECT

THE ACME COMPANY INC.
20 First St.
Somewhere, WV XXXXX

RE: _____ Asphalt Items QC Plan

Project CID#: ########
Fed/State Project #: ####- ## - ####.##
Description: Falling Slide
County : XXXXXXX

Dear Mr./Ms/Mrs. ___________________,

Your request to use your Master Asphalt Items Quality Control Plan **(M# - ######)** for Asphalt Items on the project referenced above, has been reviewed and found to be acceptable for the following items:

As work progresses throughout this project an addendum(s) may be required to this QCP to keep the QC program current. **Please use M# - ###### when corresponding about this QC plan. Also note that personnel may be required to show proof of certification for testing.** Please make sure that all appropriate personnel have a copy of this plan in their possession.

For Division/District use

The Master Quality Control Plan can be reviewed in ProjectWise at this Link:

WVDOT ORG>D0#>year>MASTER QC PLANS>Contractors or Plant>Contractor Name>Name of Quality Control Plan

Very Truly Yours,

____
Title

WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
CONTRACT ADMINISTRATION DIVISION

MATERIALS PROCEDURE

COMPACTION TESTING OF HOT-MIX ASPHALT PAVEMENTS

1.0 PURPOSE

1.1 The purpose of this procedure is to establish the test methods for quality control testing by the Contractor and verification testing by the Division.

2.0 SCOPE

2.1 This procedure is applicable for all items of hot-mix asphalt pavements requiring compaction testing.

3.0 DEFINITIONS

3.1 Quality Control Testing – Testing conducted by the Contractor to monitor and control the production of their product.

3.2 Verification Testing – Testing conducted by the Division to determine specification compliance.

4.0 APPLICABLE DOCUMENTS

AASHTO R11
MP 712.21.26

5.0 EQUIPMENT

5.1 Nuclear density gauges of the backscatter type.

5.2 One measuring tape of approximately 50 feet (20 meters).

5.3 Lime or other suitable material to mark test sites.

5.4 Dry mortar sand.

5.5 Supply of T401 or T407 data sheets.

6.0 ROUNDING OF DATA

6.1 Test data must be rounded according to AASHTO R11.

6.2 Test data and calculations are rounded to the following nearest significant digit.

Station Number	1 ft (0.1 m)
Offset	1 ft (0.1 m)
Wet Density	0.1 lb/ft^3 (1 kg/m^3)
Target Density	0.1 lb/ft^3 (1 kg/m^3)
Lift Thickness Compacted	0.25 inch (1 mm)
Relative Density	1 %
Average Relative Density	1 %
Average Wet Density	0.1 lb/ft^3 (1 kg/m^3)

7.0 STANDARDIZATION OF NUCLEAR GAUGE

7.1 Warm up the gauge in accordance with the manufacturer's recommendations.

7.2 Standardization must be performed away from metal and other objects.

7.3 Clean the top of the standard block and the bottom of the gauge with a cloth.

7.4 Make sure the gauge is turned the correct way on the block.

7.5 After making the necessary adjustments on the gauge for standardization, take a four minute count for density.

7.6 Compare the standard count to the manufacturer's standard count. The standard count must be within ± 2% from the manufacturer's standard.

7.7 If the gauge is not within the specified tolerance, repeat the standardization. If the gauge will not standardize after four attempts, there is probably something wrong with the gauge. There may be electronics problems, the gauge needs calibrated, or a stability check needs to be performed. Do not use a gauge for testing if it will not standardize.

7.8 A gauge must be standardized before testing and at least every four hours during testing.

8.0 COMPARISON OF GAUGES

8.1 The gauge used for the Contractor's quality control testing should be compared with the gauge used for the Division's verification testing.

8.2 Standardize both gauges according to 7.1 through 7.8.

8.3 Place the aluminum plate provided by the Division on the standard block used for verification testing. Place the standard block on material weighing a minimum of 110 lb/ft^3 (1762 kg/m^3). The block must not be near metal or other objects during testing and must not be moved. Keep the gauges separated a minimum of 30 feet (9.1 meters) during testing.

8.4 Take 5 one minute wet density readings with each gauge in the backscatter position. The gauges are to be oriented on the block the same as for standardization.

8.5 Record the wet density readings exactly as shown on the gauge. The range of the five readings shall not exceed 1.5 lb/ft^3 (24 kg/m^3). If the readings exceed this range, perform a new set of five readings. A gauge should not be used if the repeatability of the gauge is not within this range.

8.6 Average the five readings for each gauge. The gauges are considered similar if the averages of the readings are within 3 lb/ft^3 (48 kg/m^3).

8.7 The density readings for verification testing will not be adjusted to compensate for any differences in readings between gauges.

9.0 QUALITY CONTROL TESTING

9.1 Record the test data on a T401 form.

9.2 Divide the LOT into five equal sublots.

9.3 Randomly locate a test site within each sublot according to MP 712.21.26.

9.4 Check each test site to determine if there are surface voids. Fill the voids with dry mortar sand. Avoid a build-up of fines on the surface to no more than 0.1 inch (3 mm).

9.5 Take a one minute wet density reading on each test site.

9.6 Perform the calculations on the Division approved form.

9.7 Compare the relative densities to the specification requirements.

9.8 The results of the quality control tests should be used by the Contractor to judge if the LOT will meet specifications when verification tests are performed by the Division. Corrective measures are to be taken to bring the LOT into specifications if the quality control tests indicate that a nonconformance situation exists.

10.0 LOT-BY-LOT DIVISION VERIFICATION TESTING

10.1 Once the Contractor offers a LOT of material to the Division for testing, verification testing will be performed to determine compliance to the specifications.

10.2 Randomly locate a test site within the LOT according to MP 712.21.26.

10.3 Check each test site to determine if there are surface voids. Fill the voids with dry mortar sand. Avoid a build-up of fines on the surface to no more than 0.1 inch (3 mm).

10.4 Take a one minute wet density reading in the backscatter position.

10.5 Perform the calculations on the T401 form.

10.6 Compare the percent relative density to the specification range. If the value is within the range, the LOT is accepted for density.

10.7 When the percent relative density is outside the specification range, divide the LOT into five equal sublots and randomly locate a test site in each sublot according to MP 712.21.26.

10.8 Take a wet density reading at each test site.

10.9 Average the five wet densities.

10.10 Calculate the percent relative density.

10.11 The LOT would be acceptable if the average relative density falls within the specification range. A nonconformance situation exists if the value is outside the range.

11.0 ROLLERPASS COMPACTION PROCEDURE

11.1 When the total new pavement thickness is limited, the specifications may require that compaction testing will be performed in accordance with the following rollerpass procedure.

11.2 At the beginning of the work, a test section shall be constructed with a length of 100 feet (30 meters) and the width of the paving operation except in restricted areas. If the 100 feet (30 meters) length cannot be obtained, then the test section shall be the maximum obtainable length.

11.3 If there is a concern that the existing pavement conditions may cause difficulty in obtaining the specified density requirement then the Division will either monitor or conduct density testing of the existing pavement before the test section is constructed. Five randomly located wet density tests will be conducted within the test section area and the results will be recorded on a T401 form. Additional testing may also be conducted on other sections of the existing pavement if it is considered necessary for later evaluation.

11.4 To determine the number of roller passes for lift thicknesses of less than 1.5 inches (38 mm), immediately after placement start the rolling operation on the test section and continue this process until the mat temperature reaches 175 °F (80 °C). If the mat begins to show signs of distress (such as excessive surface aggregate breakage or mat cracking) before reaching 175 °F (80 °C), then discontinue rolling and record the number of roller passes completed before the stress signs occurred. The mat temperature may be lowered to 165 °F (74 °C) if the contractor can demonstrate through the test section that additional densification can be achieved at this lower temperature without causing any pavement distress.

11.5 If the lift thickness is 1.5 inches (38 mm) or greater, the rolling operation may be stopped at 200 °F (93 °C) to conduct density testing as per Section 11.7. If additional rolling is needed then continue as per Section 11.4. If the air temperature is below 60 °F (16 °C), the rolling operation should not be halted until the mat temperature reaches 175 °F (80 °C) unless the distress signs described in Section 11.4 occur. Project conditions may require the Engineer to determine the proper rolling application for lift thicknesses of 1.5 inches (38 mm) or greater.

11.6 The Division will either conduct or closely monitor all density testing on the test section.

11.7 Divide the test section into two equal sublots and randomly locate a test site within each according to MP 712.21.26. Take a wet density reading on each sublot using the procedure described in Section 10.3 and 10.4. Determine the average wet density obtained from the two sublots and use this average to calculate the relative density of the test section. Record all rollerpass density test data on a T407 form.

11.8 If the relative density of the test section is within 92 – 96 % of the maximum density of the approved mix design, or the maximum density established by the most recent plant mix formula verification, then density has been achieved and the number of roller passes has been established for the remainder of the project.

11.9 If the relative density of the test section is above 96 % the Division will make a visual evaluation of the mat and the mixture to look for any appearance of excessive asphalt or an extremely fine mix which may result in over compaction. A review of any density test results obtained from the existing pavement will be made to determine if the existing pavement density was significantly higher than the target density of the mix. The Division will determine whether additional test sections are needed or that the pavement is compacted to the satisfaction of the Engineer with the established number of roller passes. If it is later determined, through the Contractor's daily quality control testing, that the mix had an air void content below 2.5% then proper adjustments shall be made to the mix to bring the air voids back into the allowable tolerance limits. The Division may require the Contractor to establish a new test section if such mix adjustments are required.

11.10 If the relative density of the test section is below 92 %, then a new test section shall be established and the Contractor shall make adjustments to his rolling operation in an attempt to achieve a higher density level before the mat temperature reaches 175 °F (80 °C).

11.11 If the density requirement is not met after two consecutive test sections are completed, the Division will determine whether additional test sections are needed or that the pavement is compacted to the satisfaction of the Engineer with the established number of roller passes. To help with this decision, an evaluation will be made of the existing pavement condition and any density test results obtained prior to construction of the test section will be reviewed. If it is later determined, through the Contractor's daily quality control testing, that the mix had an air void content above 5.5% then proper adjustments shall be made to the mix to bring the air voids back into the allowable tolerance limits. The Division may require the Contractor to establish a new test section if such mix adjustments are required.

11.12 The established number of roller passes shall continue for the remainder of the project unless the Division determines that weather conditions or changes in the condition of the existing roadway are affecting the rolling operation. Under such circumstances, the Division may request that a new roller pattern be established through a new test section.

11.13 The designated number of roller passes shall continue to be completed before the mat temperature falls below 175 °F (80 °C) unless the conditions of Section 11.4 have been established.

11.14 The Contractor shall designate a person to monitor and document the number of roller passes and the mat temperature through the duration of the project.

Robert K. Tinney

Robert K. Tinney, Director
Contract Administration Division

WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS AND TESTING DIVISION

MATERIALS PROCEDURE

AGGREGATE SAMPLING PROCEDURES

1. PURPOSE

1.1 To provide a uniform procedure for obtaining aggregate samples.

2. SCOPE

2.1 This procedure shall apply to the following:

(a) Process Control sampling by the Contractor.
(b) Acceptance Sampling by the Division.
(c) Independent Assurance Sampling by the Division.
(d) Record Sampling by the Division.

3. GENERAL

3.1 Taking a good sample is just as important as conducting a good test. The sampler must use every precaution to obtain samples that will show the true nature and condition of the material they represent.

3.2 Most aggregates are mixtures of various particle sizes, which tend to separate, or segregate, during transporting or stockpiling. For this reason, aggregate samples should be obtained at the last practical point before the material is incorporated into the finished product or before compaction.

3.3 Frequency of sampling will be in accordance with the applicable directives for the type sample being procured.

4. DEFINITION OF TERMS

Sampling Unit

Lot

Increments

Sublot

Sampling Unit

Field Sample

Test Portion

4.1 Lot: The quantity of material represented by an average test value, not to exceed five individual test values, calculated in accordance with MP 300.00.51.

4.2 Sublot: The quantity of material represented by a single test value. In the case where only one sample is needed for the total plan quantity, the sublot may be considered the Lot.

4.3 Sampling Unit: The quantity of material within the sublot from which increments are obtained to be combined into a field sample.

4.4 Increment: The portion of material removed from the sampling unit to be combined into a field sample.

4.5 Field Sample: A composite of increments.

4.6 Test Portion: The material split from the field sample to be used in performing a specific test.

4.7 Random Location: A location whose position depends entirely on chance. In other words, one location has as good a chance being selected as any other.

5. CONTRACTOR RESPONSIBILITY

5.1 The Contractor shall provide all reasonable facilities and furnish the Division the information, assistance and samples required by the Engineer and Inspector for proper inspecting or testing of materials and workmanship.

5.2 All materials and each part or detail of the work shall be subject to inspection by the Engineer. The Engineer or a representative shall be allowed access to all parts of the work and shall be furnished with such information and assistance by the

Contractor as is required to make a complete and detailed inspection. To facilitate the inspection of materials, all delivery tickets shall contain as a minimum the information required in MP 700.00.01.

6. SAMPLING PROCEDURES

6.1 There are four general areas from which aggregate samples are usually obtained. These include (1) Sampling from the roadway after the aggregate has been placed, but prior to compaction, (2) Sampling from a conveyor belt, (3) Sampling from a flowing stream of aggregate, and (4) Sampling from stockpiles.

6.1.1 Sampling from the roadway (e.g., bases and subbases)

6.1.1.1 The first step in obtaining a roadway sample is to locate the sublot. This is usually the quantity of material that will be represented by the one sample and is defined as a section of roadway of given width and length.

6.1.1.2 The next step is to randomly locate a sampling unit within the sublot. A sampling unit is defined as an area having dimensions of approximately12 feet by 12 feet, or an area of approximately 144 square feet in locations having any dimension less than 12 feet. Locating the sampling unit is accomplished by use of random numbers contained in Attachment I. Any pair of numbers (decimals) may be used to locate a sampling unit within the sublot. To locate the sampling unit in a sublot defined by area, the length of the area, in feet, is multiplied by one decimal of the pair, and width is multiplied by the other decimal. The resulting distances are to be measured from one end and one side of the area.

6.1.1.3 For example, a sublot of material consists of base course aggregate 26 feet wide from Station 956+00 to 965+00. The total length in feet is thus 96,500 minus 95,600, or 900 feet. A pencil tossed on Attachment I points to the pair of decimals 0.115 and 0.447. To locate the sampling unit, the first decimal is multiplied by the length of the sublot, and the companion decimal is multiplied by the width. The length value will be measured from Station 956+00 and the width value from the left-hand edge of the base. Thus, 900 multiplied 0.115 equals 104, and 26 multiplied by 0.447 equals 12, so the sampling unit would be located at Station 956+00 plus 1+04, or 957+04 at 12 feet from the left edge. This point could define the center or any corner of the sampling unit. If we use the center, a 12-foot by 12-foot sampling unit would fall between Stations 956+98 and 957+l0 with longitudinal boundaries at 6 feet and 18 feet from the left edge of the base.

6.1.1.4 Five approximately equal increments are then located within the sampling unit. This is also best accomplished by means of the random numbers in Attachment 1. Procedures to follow are essentially the same as those set forth for locating the sample unit. The five increments are taken from the sampling unit and combined to form a field sample whose weight equals or exceeds the minimum recommended in Table I. All increments shall be taken from the roadway for the full depth of the material being sampled, care being taken to exclude any foreign material which may have been incorporated during the normal construction process. The specific areas from which each increment is to be removed shall be clearly marked; a metal template placed over the area is a definite aid in securing approximately equal increment weights.

6.1.2 Sampling from a Conveyor Belt

6.1.2.1 The first step in obtaining a sample from the conveyor belt is to define the sublot. This is generally defined as a unit of time, i.e., a half-day or a day's production. The next step is to randomly locate a sampling unit within the sublot. A sampling unit in this case is generally considered to be the material contained within the length of the conveyor. Locating the sampling unit is accomplished by use of the random numbers contained in Attachment I. Any number may be used to locate a sampling unit within the sublot. To locate the sampling unit in a sublot defined by time, the length of time, usually in minutes, is multiplied by the random decimal obtained from Attachment I.

6.1.2.2 For example, a sublot of material consists of concrete aggregate used in a half-day's production estimated to be between 8:00 a.m. and 12:00 noon. A pencil tossed on Attachment I points to decimal 0.279. Thus, the sampling unit would be located somewhere within the four-hour period (8:00 a.m. to 12:00 noon). Four (hours) multiplied by 60 (minutes per hour) multiplied by 0.279 equals 67 minutes. The sampling unit would be located 67 minutes after the 8:00 a.m. startup; or at 9:07 a.m. (or as soon thereafter as practical).

6.1.2.3 Five randomly located, approximately equal increments are obtained from the sampling unit and combined to form a field sample whose weight equals or exceeds the minimum recommended in Table I. The location of the five increments is determined by multiplying the length of the belt by five random numbers. The conveyor belt is stopped while the increments are being obtained. Two templates, conforming to the shape of the belt and spaced such that the material contained between them will yield an increment of the required size, are inserted into the aggregate stream on the belt. All material between the templates is carefully scooped into a suitable container, including all fines on the belt collected with a brush and dustpan.

6.1.3 Sampling from a Flowing Aggregate Stream (bin or belt discharge)

6.1.3.1 Definition of the sublot and location of the sampling unit is generally identical with sampling from a conveyor belt, with the exception that the sampling unit in this case is defined as that material which will flow during a five-minute period. Once the sampling unit is located, five approximately equal increments, randomly spaced, are obtained and combined to form a field sample whose weight equals or exceeds the minimum recommended in Table I. Each increment is taken from the entire cross-section of the material as it is being discharged. It is usually necessary to have a special device constructed for use at each individual plant. This device will consist of a pan of sufficient size to intercept the entire cross-section of the discharge stream and hold the required quantity of material. A set of rails may be necessary to support the pan as it is passed under the discharge stream. If the sampling pan overflows, it should be struck level so that only material that is within the pan is retained.

6.1.4 Sampling from a Stockpile

6.1.4.1 If possible, stockpile sampling should be avoided when sampling to determine the gradation of an aggregate. However, circumstances sometimes make it necessary, and when this occurs a sampling plan and the number of samples to be taken must be considered for each specific case. Stockpiled aggregates tend to segregate with the coarser particles rolling to the outside base of the pile, which makes gradation representation difficult. Because of this, every effort should be made to enlist the services of power equipment (such as a front-end loader) to develop a separate small sampling pile composed of material taken from various levels and depths in the main stockpile. Increments from this pile may be combined, thoroughly mixed, and reduced by quartering and/or sample splitter to obtain the field sample. Methods for quartering and splitting samples are given in AASHTO R76. If power equipment is not available, hand sampling may be employed to obtain at least three increments per sample: One increment taken from the top one third of the pile, one from the middle and one from the bottom third. When hand sampling, the outer layer of the pile should be removed (scraped away with the shovel) at the point prior to sampling.

7. WEIGHTS REQUIRED

7.1 Field Sample Weights

7.1.1 Field sample weights as listed in Table I are minimum values. Actual weights required must be predicated on the type and number of tests to which the material is to be subjected. The amounts specified in Table I will provide adequate material for routine gradation and quality analysis.

7.2 Test Portion Weight

7.2.1 The weight of the test portion to be obtained from the field sample for a specific test will be defined in the Standard Procedures of the test involved. Reduction of the field sample into test portions is done with a sample splitter. The weight of test portion recommended for gradation testing is given in Table II.

8. TRANSPORTING SAMPLES

8.1 Testing at Site of Sampling

8.1.1 Samples taken for testing in the field may be placed in any suitable clean container of appropriate size which is secure enough to prevent loss of material when transporting the sample to the testing location.

8.2 Samples to be Shipped from Site of Sampling

8.2.1 Samples to be shipped should be placed in standard sampling sacks. If the sample contains an appreciable quantity of fine material, a plastic liner should be put in the sack to prevent loss of the fines. Each sack must be securely tied to prevent loss of material in transit. It is also essential that sample identification be maintained from the field to the testing site. Each sack must have appropriate indelible identification attached and enclosed so that field reporting, laboratory logging, and test reporting may be facilitated.

12/08/2021

Ronald L. Stanevich, P.E.
Director
Materials Control, Soils and Testing Division

RLS:Ms
Attachments

TABLE I

WEIGHT OF SAMPLES

NOMINAL MAXIMUM SIZE OF PARTICLES*	MINIMUM WEIGHT OF FIELD SAMPLES	
Sieve Size	Kilo	lb
No. 8	10	25
No. 4	10	25
3/8 in.	10	25
1/2 in.	15	35
3/4 in.	25	55
1 in.	50	110
1 1/2 in.	75	165
2 in.	100	220
2 1/2 in.	125	275
3 in.	150	330
3 1/2 in.	175	385

*The nominal maximum size of particles is defined as the largest sieve size listed in the applicable specifications upon which any material is permitted to be retained. Exception: If the specification tolerances are such that no sieve listed has a range of X-100 percent passing, then the next smallest standard sieve, as listed in Table I, and below that sieve which 100 percent must past will be considered the nominal maximum size.

TABLE II

TEST PORTION FOR GRADATION

NOMINAL MAXIMUM SIZE OF PARTICLES	MINIMUM WEIGHT OF TEST PORTION	
Sieve Size	Kilo	lb
No. 8	0.1	0.3
No. 4	0.5	1.0
3/8 in.	1.0	2.0
1/2 in.	2.0	4.0
3/4 in.	5.0	11.0
1 in.	10.0	22.0
1 1/2 in.	15.0	33.0
2 in.	20.0	44.0
2 1/2 in.	35.0	77.0
3 in.	60.0	130.0
3 1/2 in.	100.0	220.0

ATTACHMENT I

RANDOM NUMBERS

.858	.082	.886	.125	.263	.176	.551	.711	.355	.698
.576	.417	.242	.316	.960	.879	.444	.323	.331	.179
.587	.288	.835	.636	.596	.174	.866	.685	.066	.170
.068	.391	.139	.002	.159	.423	.629	.631	.979	.399
.140	.324	.215	.358	.663	.193	.215	.667	.627	.595
.574	.601	.623	.855	.339	.486	.065	.627	.458	.137
.966	.589	.751	.308	.025	.836	.200	.055	.510	.656
.608	.910	.944	.281	.539	.371	.217	.882	.324	.284
.215	.355	.645	.450	.719	.057	.287	.146	.135	.903
.761	.883	.711	.388	.928	.654	.815	.570	.539	.600
.869	.222	.115	.447	.658	.989	.921	.924	.560	.447
.562	.036	.302	.673	.911	.512	.972	.576	.838	.014
.481	.791	.454	.731	.770	.500	.980	.183	.385	.012
.599	.966	.356	.183	.797	.503	.180	.657	.077	.165
.464	.747	.299	.530	.675	.646	.385	.109	.780	.699
.675	.654	.221	.777	.172	.738	.324	.669	.079	.587
.269	.707	.372	.486	.340	.680	.928	.397	.337	.564
.338	.917	.942	.985	.838	.805	.278	.898	.906	.939
.130	.575	.195	.887	.142	.488	.316	.935	.403	.629
.011	.283	.762	.988	.102	.068	.902	.850	.569	.977
.683	.441	.572	.486	.732	.721	.275	.023	.088	.402
.493	.155	.530	.125	.841	.171	.794	.850	.797	.367
.059	.502	.963	.055	.128	.655	.043	.293	.792	.739
.996	.729	.370	.139	.306	.858	.183	.464	.457	.863
.240	.972	.495	.696	.350	.642	.188	.135	.470	.765

WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS AND TESTING DIVISION

MATERIALS PROCEDURE

NUCLEAR DENSITY TEST BY THE ROLLER PASS METHOD

1. PURPOSE

1.1 The purpose of this procedure is to determine the density of construction materials by the roller pass method. The procedure consists of two parts, with Part I to determine the required maximum density and Part II to compare field densities to the required maximum density.

2. SCOPE

2.1 This test method or method of testing is applicable to aggregate base courses, select material for backfilling, crushed aggregate backfills, granular material, subgrade, and random material having 40% or more of +3/4-inch material as specified in MP 717.04.21.

3. REFERENCES

3.1 MP 712.21.26

3.2 MP 717.04.21

4. EQUIPMENT

4.1 One complete nuclear density gauge unit meeting the requirements specified in MP 717.04.21. This would include the manufacturer's printout of standard counts.

4.2 One measuring tape, approximately 50 feet.

4.3 Lime, chalk, lumber crayon, or other suitable material to mark test sites

4.4 Dry silica sand

4.5 Supply of data sheets

4.6 One vehicle meeting the safety and security requirements of the Nuclear Regulatory Commission for transporting nuclear density gauges.

5. PERSONNEL TRAINING

5.1 All personnel performing the testing must meet the minimum training requirements specified in MP 717.04.21.

5.2 All personnel must know and follow the requirements of the Nuclear Regulatory Commission.

6. ROUNDING OF DATA

6.1 Test values and calculations are to be rounded according to the following procedure:

6.1.1 If the figure following the last significant number to be retained is five or larger, increase the last significant number to be retained by one.

6.1.2 If the figure following the last significant number to be retained is less than five, the last significant number is left unchanged.

6.2 Test values and calculations shall be rounded to the following nearest significant digit:

6.2.1 Form T-313[1] (Test Section)

Lift Thickness Compacted	0.1 in.
Depth Below Grade	1 ft
Length of Test Section	1 ft
Width of Test Section	1 ft
Station Number	1 ft
Offset	1 ft
Dry Density (DA)	1 PCF
Average Density (DB)	1 PCF
Maximum Density (DC)	1 PCF

6.2.2 Form T-317[1] (Quality Control Tests)

Station Number	1 ft
Offset	1 ft
Depth Below Grade	1 ft
Lift Thickness Compacted	0.1 in.
Maximum Density (dc)	1 PCF
Dry Density (de)	1 PCF
Relative Density (df)	1%
Average df (x)	0.1%
Target (f)	1%
Quality Index (ql)	0.01
Within Tolerance (dg)	1%
Minimum Percent for 100% pay (dh)	1%

7. PREPARATION FOR TESTING

7.1 Standardization of the Nuclear Gauge

[1] https://transportation.wv.gov/highways/mcst/Pages/tbox.aspx

7.1.1 Warm up the gauge according to the manufacturer's recommendations.

7.1.2 Standardization of the gauge must be performed away from metal and other objects.

7.1.3 Clean the top of the standard block and the bottom of the gauge with a cloth.

7.1.4 Standardize according to manufacturer's recommendations.

7.1.5 Compare the standard counts to the manufacturer's standard counts using tolerances acceptable to the Division. For the Troxler 3400 series gauge, the standard counts must be within ± 2% for density and ± 4% for moisture from the manufacturer's standards.

7.1.6 If the gauge is not within the specified tolerances for either moisture or density, repeat section 7.1.4 -7.1.5. If the gauge will not standardize for either moisture or density after 4 attempts, a different gauge shall be used. The gauge which failed to standardize may be used again in the future if the procedure referenced in Section 5.2.10 of MP 717.04.21 is followed and the gauge is found to be stable.

7.1.7 Gauges shall be standardized before testing and at least every four hours during testing.

7.1.8 When a gauge is to be used for testing pipe or structure backfill in a trench, first check the standardization of the gauge according to sections 7.1 - 7.1.5. If the gauge is functioning properly, standardize the gauge in the trench. The standard counts in the trench shall be used for testing in the trench only and the tolerances would not be applied to the standard counts taken in the trench. When the gauge is moved to a non-trench condition for testing, new standard counts shall be required.

8. PART I PROCEDURE FOR DETERMINING THE MAXIMUM DENSITY

8.1 All data and calculations for Part I of this procedure will be recorded on form T-313 (copy attached). Record the Contract ID, project number, lab number etc. before starting the test.

8.2 The test is to be performed at the beginning of placement of an item. However, any problems with the material, placement, or compaction equipment shall be corrected prior to performing the test.

8.3 The test section will be 100 feet long by the width being placed in one operation, except in restricted areas.

8.3.1 In restricted areas, where the 100-foot length cannot be obtained, check the project's records to determine if a maximum density for the material has been determined on the project. If the material, lift thickness, and compaction equipment remain unchanged, the existing maximum density shall be used for Part II of this procedure, if available. A maximum density determined in a restricted area shall not be used in a non-restricted area. If a maximum density is not available for the material, obtain the largest test section possible. For pipe backfill, a lift on both sides of the pipe can be used.

8.4 Divide the test section into 5 equal subsections and number the subsections. Randomly locate a test site within each of the subsections according to MP 712.21.26.

8.5 Water shall be added to untreated aggregates, if necessary, in a quantity satisfactory to the Engineer. The aggregate must visually appear wet to properly compact.

8.6 Once the material had been placed in the test section, the material shall be rolled with compaction equipment meeting the following requirements:

8.6.1 All compaction equipment must be in good working condition.

8.6.2 The materials shall be compacted with rollers providing a minimum applied force of 10 tons.

8.6.3 In restricted areas, inaccessible to conventional rollers, the compaction equipment must be satisfactory to the Engineer to provide the desired compactive effort. The Division may request verification that the above compaction equipment meets the specified requirements.

8.7 The test section shall be rolled with 12 roller passes. A roller pass is one complete coverage over the material. In restricted areas, where conventional rollers cannot be used, the material shall be compacted until it appears well densified or to the satisfaction of the Engineer.

8.8 If the material shears or breaks down during rolling, the number of roller passes may need to be reduced. The designated number of roller passes must not be changed without the approval of the Engineer.

8.9 Once the material has been rolled, testing will be performed on test sites 1 and 2.

8.10 Smooth and level the test site. Fill any voids with fines scraped from the surface, but no more than 1/8 inch.

8.10.1 Place the guide plate on the test site. Next place the drive rod in the guide plate and while standing on the plate, drive the rod at least two inches deeper than the location where the end of the gauge source rod will be when testing. The gauge source rod can be extended in two-inch (50 mm) increments. The source rod must be as deep as possible within the lift but must not extend beyond the lift. For example, a five-inch lift would be tested with the source rod in the four-inch position and the hole would be six inches deep. Carefully remove the drive rod to prevent material from falling into the hole.

8.10.2 Place the gauge over the test site and insert the source rod to the desired depth. Pull the gauge tight against the side of the hole toward the scaler. Make sure the gauge is sitting flush on the material. Mark the outline of the gauge with lime or other suitable material so the test sites can be relocated.

8.10.3 Take a one-minute density reading.

8.10.3.1 Record the dry density (DA) in Section A of form T-313. Perform the same testing on site 2.

8.11 Average the two dry densities (DA) obtained in section 8.10.3.1.

8.12 Roll the material in the test section two additional roller passes. In restricted areas, the compaction equipment would pass over the material the above indicated number of passes.

8.13 After the material has been rolled the additional number of passes, perform tests again on sites 1 and 2 according to sections 8.10 through 8.10.3 and record the values in section B.

8.14 Average the two densities according to section 8.11.

8.15 Compare the value in section 8.14 to the value obtained in section 8.11. If the increase in density is 1 PCF or less, the material is considered to have achieved its maximum density. If the increase in density is greater than 1 PCF, roll the material two additional passes according to section 8.12 and repeat the testing on sites 1 and 2. Continue the rolling and testing sequence until the increase in density between two consecutive rolling sequences is 1 PCF or less. The Division may direct the contractor to cease rolling even though the increase is more than 1 PCF if the material is breaking down.

8.16 Once the increase in density is 1 PCF or less, move the last two density readings to the maximum density determination section on form T-313. Take density measurements on sites 3, 4, and 5.

8.17 The average of the five density readings is the maximum density (DC) for this material.

8.17.1 The maximum density will be used to control the material for Part II of this procedure.

8.17.2 Division personnel may request that Part I be repeated if the test was not performed properly or the maximum density obtained does not appear to be realistic.

9. PART II QUALITY CONTROL TESTING

9.1 All test data and calculations for Part II of this procedure will be recorded on form T-317 (copy attached). Record the Contract ID, project number, item number, etc. on the form before starting the testing.

9.2 The lot number would have a prefix letter based on the following designations for the use of the material being tested:

Embankment:	F
Subgrade:	S
Base:	B
Pipe/Structure Backfill:	p

9.3 Transfer the maximum density (DC) and the lab number from form T-313 to form T-317. Record the lab number in the section for reference lab number.

9.4 Randomly locate the test site according to MP 712.21.26.

9.5 Determine the dry density (DE) with the nuclear gauge according to the procedure described in sections 8.10 through 8.10.3. The test sites do not have to be marked on the roadway.

9.6 Calculate the percent relative density (DF) by using the equation on form T-317.

9.7 Perform the remaining four tests in the lot. Five tests are always required to evaluate a lot.

9.8 Calculate the average relative density *(X)* for the five tests in the lot.

9.9 Obtain the target percentage of dry density (T) from the project's governing specifications.

9.10 Determine the range (R) of the relative densities (DF) by subtracting the smallest value from the largest.

9.11 Calculate the quality index (QL) by using the equation on form T-317.

9.12 Use the Table for Estimating the Percent of a Lot Within Tolerance (copy attached) and determine the percent within tolerance (DG) that corresponds to the QL value calculated in section 9.11 above.

9.13 Obtain the minimum percent for 100% pay (DH) from the project's governing specifications.

9.14 For a lot to meet specifications, the percent within tolerance (DG) must be equal to or greater than the minimum percent for 100% pay (DH).

10. GENERAL

10.1 Independent tests for similarity checks can be recorded on form T-317. Use only the applicable sections of the form.

10.2 If the material changes or the material is supplied from a new source, repeat Part I to obtain new control data.

10.3 If the percent relative densities are consistently above 105 percent or below 95 percent, and there is no apparent cause for the high or low values, repeat Part I to obtain new control data.

10.4 Test data for several lots can be recorded on form T-317.

05/31/2023

Ronald L. Stanevich, PE
Director
Materials Control, Soils & Testing Division

MP 700.00.24 Steward – Aggregate and Soils Section
RLS:R

ATTACHMENTS

TABLE FOR ESTIMATING PERCENT OF LOT WITHIN TOLERANCE

Quality Index (QL) Positive Values	Percent Within Tolerance
.66	99
.65	98
.62	97
.60	96
.58	95
.57	94
.55	93
.53	92
.51	91
.50	90
.48	89
.46	88
.45	87
.44	86
.42	85
.41	84
.40	83
.38	82
.37	81
.36	80
.34	79
.33	78
.32	77
.30	76
.29	75
.28	74
.27	73
.25	72
.24	71
.23	70
.22	69
.21	68
.19	67
.18	66
.17	65
.16	64
.15	63
.14	62
.13	61
.11	60
.10	59
.09	58
.08	57
.07	56
.06	55
.05	54
.04	53
.02	52
.01	51
.00	50

Quality Index (QL) Negative Values	Percent Within Tolerance
.00	50
.01	49
.02	48
.04	47
.05	46
.06	45
.07	44
.08	43
.09	42
.10	41
.11	40
.13	39
.14	38
.15	37
.16	36
.17	35
.18	34
.19	33
.21	32
.22	31
.23	30
.24	29
.25	28
.27	27
.28	26
.29	25
.30	24
.32	23
.33	22
.34	21
.36	20
.37	19
.38	18
.40	17
.41	16
.42	15
.44	14
.45	13
.46	12
.48	11
.50	10
.51	9
.53	8
.55	7
.57	6
.58	5
.60	4
.62	3
.63	2
.66	1

West Virginia Division of Highways
Materials Control Soil and Testing Division

FORM T-313
MP 700.00.24
REV. 09-22

Lab Number ____________
Auth. Number ____________
Project Number ____________
District Number ____________
Item Number ____________
Date ____________ Contract ID ____________

Source of Material:			Length of Test Section:	
Roller Type:			Width of Test Section:	
Roller Weight	Static:	Working:	Gauge Number	
Lift Thickness Compacted:			Manufacturer's Standards	
Depth Below Grade:			Density:	Moisture:
Depth of Gauge Source:			Standard Counts	
Observed	Yes	No	Density:	Moisture:

Test Site Number	1	2	3	4	5
Station Number					
Offset					

A	Number of Passes		
	Test Site	DA	Dry Density
	1		
	2		
DB	Average		

B	Number of Passes		
	Test Site	DA	Dry Density
	1		
	2		
DB	Average		

C	Number of Passes		
	Test Site	DA	Dry Density
	1		
	2		
DB	Average		

D	Number of Passes		
	Test Site	DA	Dry Density
	1		
	2		
DB	Average		

$DB = \sum DA / 2$

$DC = \sum DA / 5$

Maximum Density Determination		
Test Site	DA	Dry Density
1		
2		
3		
4		
5		
DC Max. Density		

Inspector's Name: ____________
Inspector's Signature: ____________

Project's Evaluation

Checked By: ____________
Date: ____________

[2] https://transportation.wv.gov/highways/mcst/Pages/tbox.aspx

WEST VIRGINIA DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS & TESTING DIVISION

LAB NUMBER ____________
AUTH NUMBER ____________
PROJECT NUMBER ____________
DISTRICT ____________
ITEM NUMBER ____________
CONTRACT ID ____________

FORM T-317
MP 700.00.24
REV. 09-22

GAUGE #	DATE					
MANUFACTURER'S DENSITY STANDARD	LOT NUMBER					
	BEGINNING STATION					
	ENDING STATION					
MANUFACTURER'S MOISTURE STANDARD	OFFSET					
	DEPTH BELOW GRADE					
	DEPTH OF GAUGE SOURCE					
	LIFT THICKNESS COMPACTED					
	DENSITY STANDARD					
	MOISTURE STANDARD					

DC FROM TEST SECTION

$$DF = \frac{DE\,(100)}{DC}$$

$$\bar{X} = \frac{\sum DF}{5}$$

$$QL = \frac{\bar{X} - T}{R}$$

	DC	MAXIMUM DENSITY					
		REFRENCE LAB NUMBER					
TEST NUMBER 1	DE	DRY DENSITY					
	DF	% RELATIVE DENSITY					
TEST NUMBER 2	DE	DRY DENSITY					
	DF	% RELATIVE DENSITY					
TEST NUMBER 3	DE	DRY DENSITY					
	DF	% RELATIVE DENSITY					
TEST NUMBER 4	DE	DRY DENSITY					
	DF	% RELATIVE DENSITY					
TEST NUMBER 5	DE	DRY DENSITY					
	DF	% RELATIVE DENSITY					
LOT EVALUATION	$\bar{X}$	AVERAGE DF					
	T	TARGET	95%	95%	95%	95%	95%
	QL	QUALITY INDEX					
	DG	% WITHIN TOLERANCE					
	DH	MIN. FOR 100% PAY	80%	80%	80%	80%	80%
	DI	PASS / FAIL					

INSPECTOR'S NAME: ____________
INSPECTOR'S SIGNATURE: ____________

PROJECT'S EVALUATION

CHECKED BY: ____________ DATE: ____________

---

[3] https://transportation.wv.gov/highways/mcst/Pages/tbox.aspx

WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS DIVISION

MATERIALS PROCEDURE

PROCEDURE FOR DETERMINING THE RANDOM LOCATION OF SOIL AND AGGREGATE COMPACTION TESTS

1. PURPOSE

1.1 This procedure provides methods for determining the random locations for soil and aggregate compaction tests on WVDOT projects.

2. SCOPE

2.1 This procedure is applicable for locating all soil and aggregate compaction tests.

3. EQUIPMENT

3.1 Measuring tape, approximately 50 feet.

4. DEFINITIONS

4.1 Test Section- A test section is an isolated quantity of material used to determine the maximum density and optimum moisture content of the material using the roller pass method.

4.2 Lot- A lot is an isolated quantity of specified material from a single source or a measured amount of specified construction assumed to be produced by the same process.

5. PROCEDURE

5.1 Compaction test site locations shall be randomly located along the roadway centerline (length) and offset (width) randomly from this reference line. Some test site locations, such as pipe backfill, require random selection of lifts for the tests and a random determination of the side of the pipe backfill to test.

5.2 Selection of random numbers:

5.2.1 Determine the number of test sites which will be required for the lot or test section.

5.2.2 The table of random numbers (Table 1 attached) or a calculator, which will generate random numbers, can be used.

5.2.3 The table of random numbers contains 5 sections with 2 columns of numbers in each section.

5.2.3.1 The first column of numbers in each section is for determining the test site along the centerline. The second column of numbers is for determining the distance from the centerline (offset). Either column of numbers can be used for selecting lifts to be tested.

5.2.3.2 To use the table, select a random point on the table by tossing a pencil upon the page or blindly pointing out a location with the finger. The selection of random numbers will consist of a pair of random numbers. Once the point is located, select the number in the first column for the length and the corresponding number in the right column for the width. When more than one pair of random numbers is needed, continue selecting the pairs of numbers down the page. If the bottom of the page is reached, go to the top of the next section to the right or to the top of the first section on the left side of the page if the bottom of the right most section of the page is reached. When selecting lifts to be tested, only single random numbers are needed and can be obtained from any of the columns of numbers.

5.2.3.3 To use a calculator, which will generate random numbers, select all numbers needed for a test site before selecting numbers for additional test sites.

5.2.3.4 Round to the nearest whole number when calculating the test site location.

5.3 Location of test sites:

5.3.1 There are many variations in the required number of tests and the physical dimensions of the area to be tested.

5.3.2 Random location of tests on a single lift that rectangular in shape (see Example 1 of Attachment).

5.3.2.1 Generally, the Materials Procedure used for testing a material and/or Specifications requires a lot, portion of a lot, or a test section to determine the maximum compacted density of a material to be divided into equal sublots or subsections when more than one test is required.

5.3.2.2 Divide the length of the area along the centerline by the number of tests to determine the length of each sublot or subsection.

5.3.2.3 From the beginning station number, add the length of the subsection or sublot to the station number to determine the station number for the beginning of the next sublot or subsection. Next add the length of the subsection or sublot to this station number to determine the station number at the beginning of the next subsection or sublot. Continue this procedure until the beginning station numbers for all subsections or sublots have been calculated.

5.3.2.4 Select the random numbers according to Section 5.2.

5.3.2.5 Multiply the length of the subsections or sublots by the random numbers selected for the length. Add the values to the corresponding station numbers for the beginning of each subsection or sublot. The station numbers locate the test sites along centerline.

5.3.2.6 Next multiply the width of the test section or lot by the random numbers selected for the offset. The offset can be calculated from the left or right side of the test area and test location designated in relation to centerline. If the test site falls on the edge of the lot or sublot, move 2 feet into the lot and perform the test at that location. Alternatively, a new set of random numbers can be used to avoid this occurrence.

5.3.2.7 When the centerline is not contained within the area to be tested, the offset distance of the lot or test section from the centerline shall be determined. This will usually be a constant value. Always calculate the offset by working from the side nearest the centerline. Add each of the values calculated in 5.3.2.7 to the constant value. The values establish the offset distance of each test site from the centerline. Designate if the offset is left or right of centerline.

5.3.3 Random location of test sites on a single lift that is irregular in shape (see Example 2 attached):

5.3.3.1 Determine the dimensions of the area to be tested.

5.3.3.2 Determine the minimum dimensions of a rectangle that will contain the area to be tested and has two sides parallel to centerline.

5.3.3.3 Divide the rectangle into the desired number of subsections or sublots and randomly locate the test sites locations as in sections 5.3.2. If a test site location falls outside the area to be tested, obtain a new set of random numbers for the test site and recalculate the test site location. Continue this procedure until the test site falls within the area to be tested.

5.3.4 Random selection of lifts to be tested (see Example 3 attached):

5.3.4.1 When testing certain materials, especially backfill material, where an area to be backfilled will constitute a lot of material to be tested, a random selection of lifts shall be tested.

5.3.4.2 Determine the projected number of lifts to be contained within the lot. Divide the number of lifts by the number of tests in the lot. If the value is not an even number, assign an additional lift to the first sublot and continue to assign a lift to each consecutive sublot until all remaining lifts have been assigned to a sublot.

5.3.4.3 By starting with the bottom lift, number the lifts in the lot, select a single random number for each test site.

5.3.4.4 Multiply each random number by the number of lifts in each sublot and round the values to whole numbers. Each value designates which lift in each sublot that will be tested.

5.3.4.5 Once the lifts to be tested have been selected, the random location of the test site on each lift can be determined.

5.3.4.6 The test site location can be found by multiplying the length of the lot by the first column of random numbers in the section. The offset of the test site location can be calculated by multiplying the second column of random numbers in the section by the width of the lot, if applicable.

5.3.5 Random selection of the side of backfill for pipe culverts:

5.3.5.1 When a lot of pipe backfill is being tested, tests shall be performed on both sides of the pipe. The side to be tested shall be randomly selected by using the random numbers selected for the location of the tests along the pipe. If the random number is less than 0.500, the test is on the left side and greater than or equal to 0.500 on the right side of the pipe.

5.3.5.2 The test site location's length is calculated by multiplying the denoted random number by the length of the lot of the pipe backfill.

Michael Mance Digitally signed by Michael Mance Date: 2025.01.02 18:53:39 -05'00'

Michael A. Mance, P.E.
Director
Materials Control, Soils & Testing Division

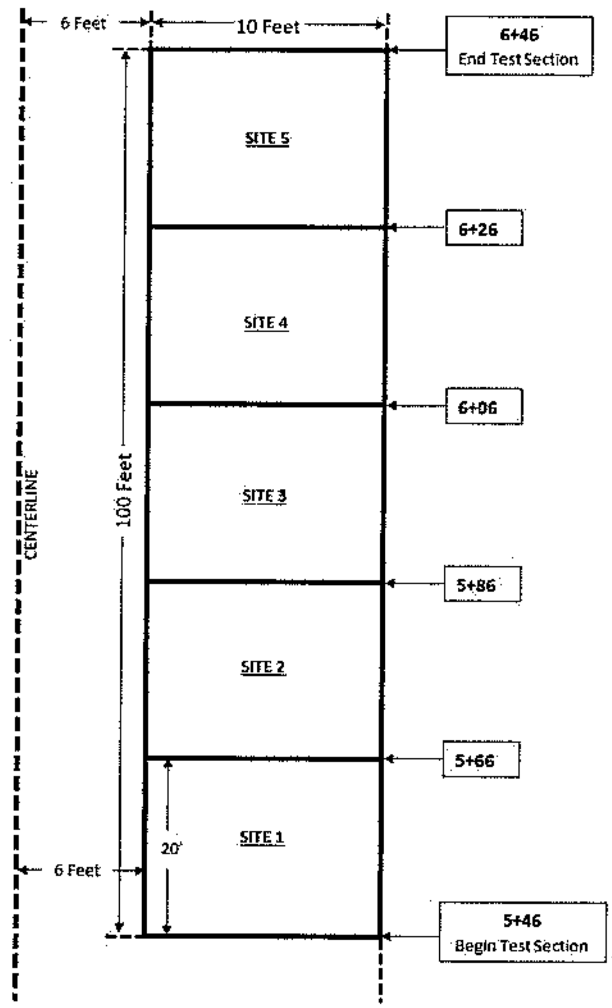
MP 712.21.26 Steward – Laboratory Support Section
MM:W
ATTACHMENTS

TABLE 1 RANDOM NUMBERS

.858	.082	.886	.125	.263	.176	.551	.711	.355	.698
.576	.417	.242	.316	.960	.819	.444	.323	.331	.179
.687	.288	.835	.636	.596	.174	.866	.685	.066	.170
.068	.391	.739	.002	.159	.423	.629	.631	.979	.399
.140	.324	.215	.358	.663	.193	.215	.667	.627	.595
.574	.601	.623	.855	.339	.486	.065	.627	.458	.137
.966	.529	.757	.308	.025	.836	.200	.055	.510	.656
.608	.910	.944	.281	.539	.371	.217	.882	.324	.284
.215	.355	.645	.460	.719	.057	.237	.146	.135	.903
.761	.883	.771	.388	.928	.654	.815	.570	.539	.600
.869	.222	.115	.447	.658	.989	.921	.924	.560	.447
.562	.036	.302	.673	.911	.512	.972	.576	.838	.014
.481	.791	.454	.731	.770	.500	.980	.183	.385	.012
.599	.966	.356	.183	.797	.503	.180	.657	.077	.165
.464	.747	.299	.530	.675	.646	.385	.109	.780	.699
.675	.654	.221	.777	.172	.738	.324	.669	.079	.587
.279	.707	.372	.486	.340	.680	.928	.397	.337	.564
.338	.917	.942	.985	.838	.805	.278	.898	.906	.939
.316	.935	.403	.629	.130	.575	.195	.887	.142	.488
.011	.283	.762	.988	.102	.068	.902	.850	.569	.977
.683	.441	.572	.486	.732	.721	.275	.023	.088	.402
.493	.155	.530	.125	.841	.171	.794	.850	.797	.367
.059	.502	.963	.055	.128	.655	.043	.293	.792	.739
.996	.729	.370	.139	.306	.858	.183	.464	.457	.863
.240	.972	.495	.696	.350	.642	.188	.135	.470	.765

EXAMPLE 1

Length of test section = 100 ft Width of section = 10 ft
Number of tests required = 5
4 equal subsections 100/5 = 20 ft
Test section starts at station 5+46



Station number at the beginning of each subsection

A. 5+46
B. 5+46 + 20 = 5+66
C. 5+66 + 20 = 5+86
D. 5+86 + 20 = 6+06
E. 6+06 + 20 = 6 +26

Random Numbers

	Length	Width
A.	.869	.222
B.	.562	.036
C.	.481	.791
D.	.599	.966
E.	.464	.747

Multiply the length of each subsection by the random numbers for the length.

A. 20 x .869 = 17
B. 20 x .562 = 11
C. 20 x .481 = 10
D. 20 x .599 = 12
E. 20 x .464 = 9

Add the values to the beginning station numbers of each subsection to determine the station number for each test.

A. 5+46 + 17 = 5+63
B. 5+66 + 11 = 5+77
C. 5+86 + 10 = 5+96
D. 6+06 + 12 = 6+18
E. 6+26 + 9 = 6 +35

Multiply the width of each subsection by the random numbers for the width.

A. 10 x .222 = 2
B. 10 x .036 = 0
C. 10 x .791 = 8
D. 10 x .966 = 10
E. 10 x .747 = 7

Add the values to the constant distance the test section is from the centerline and label the values as right of centerline .

A. 6 + 2 = 8 ft right of centerline
B. 6 + 0 = 0 ft right of centerline → Test shall be taken 8 ft right of centerline
C. 6 + 8 = 14 ft right of centerline
D. 6 + 10= 16 ft right of centerline
E. 6 + 7 = 13 ft right of centerline

EXAMPLE 2

The shaded area designates the lift to be tested. For this example, 2 sublots are required with 1 test in each sublot.

Since the area to be tested is not rectangular in shape, place the smallest rectangle around the area that will include all the shaded area.

Divide the rectangle into 2 equal areas (160 feet long by 90 feet wide).

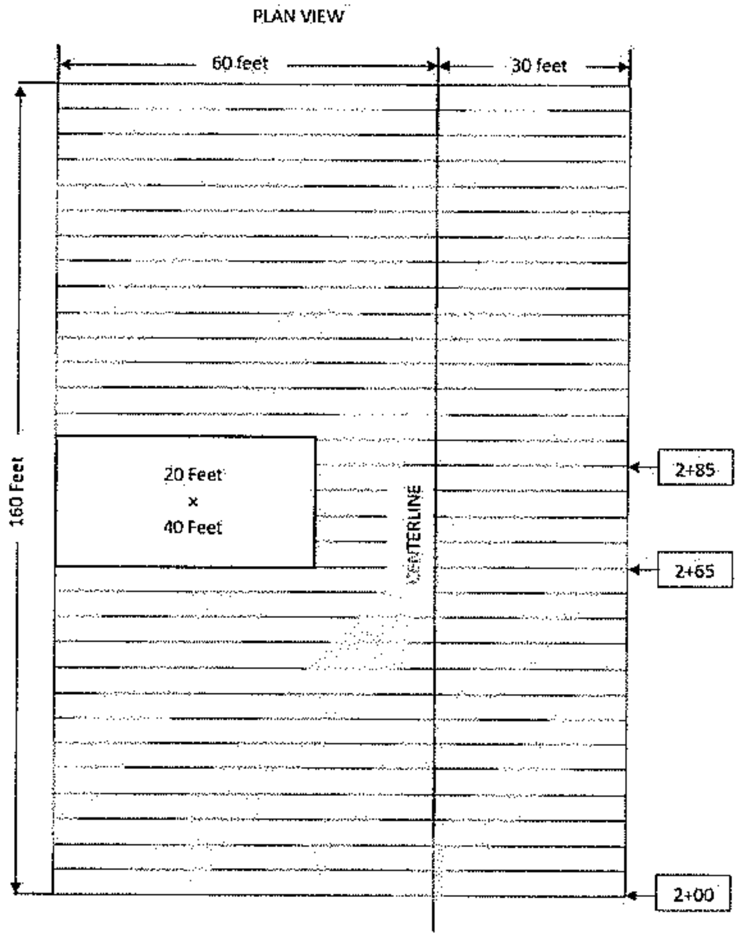
Since the centerline is located within the area to be tested, the offset can be calculated and measured from either side.
For this example, work from the right side.

Determine the station number for the beginning of each sublot.

Sublot No. 1	2+00
Sublot No. 2	2+00 + 80 = 2+80

Random Numbers
Since there is the possibility that the location of a test site may fall outside the area to be tested, an additional set of random numbers was selected.

	Length	Width
A.	.902	.850
B.	.275	.023
C.	.794	.850



Multiply the random number by the length of the sublot (80 x .902 = 72 feet). Add the value of the beginning station number (2+00 + 72 = 2+72). Multiply the width of the sublot by the random number (90 x .850 = 76 feet). By working from the right side, it is 30 feet to the centerline, therefore the test site is 76 – 30 = 46 feet to the left of centerline. The test site falls outside the test area.

By using the next set of random numbers, calculate the test site location.

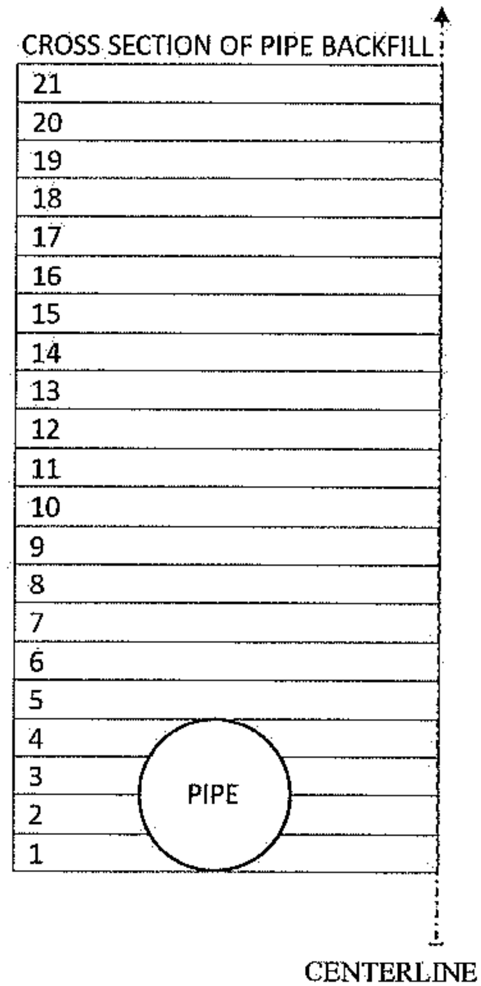
80 x .275 = 22 feet	90 x .023 = 2 feet
2+00 + 22 = 2+22	30 – 2 feet = 28 feet right of centerline

The test site for sublot 1 now falls within the test area.

Calculate the test location for sublot 2.

80 x .794 = 64 feet	90 x .850 = 76 feet
2+80 + 64 = 3+44	76 - 30 = 46 feet left of centerline

EXAMPLE 3



21 lifts of material are required to backfill the pipe.

All of the backfill material is included in 1 lot. There are 5 tests required with 1 test in each sublot.

Divide the number of lifts by the number of sublots to determine the number of lifts in each sublot (21/5 = lifts with 1 lift left over). This includes the lift in sublot number 1.

Sublot Number 1	Lifts 1 – 5
Sublot Number 2	Lifts 6 - 9
Sublot Number 3	Lifts 10 - 13
Sublot Number 4	Lifts 14 - 17
Sublot Number 5	Lifts 18 – 21

Random numbers for lift selection.

- A. .599
- B. .464
- C. .675
- D. .279
- E. .338

Multiply the number of lifts in the sublot by the random numbers. The values determine which lift in each sublot to test.

A.	5 x .599 = 3	Test lift 3 in sublot number 1, Lift number 3
B.	4 x .464 = 2	Test lift 2 in sublot number 2, Lift number 7
C.	4 x .675 = 3	Test lift 3 in sublot number 3, Lift number 12
D.	4 x .279 = 1	Test lift 1 in sublot number 4, Lift number 14
E.	4 x .338 = 1	Test lift 1 in sublot number 5, Lift number 18

Test location

	Length	Width
A.	.627	.595
B.	.458	.137
C.	.510	.656
D.	.324	.284
E.	.135	.903

Multiply the first column of numbers by the length of the sublot. Then multiply the second column by the width of the sublot. For this example, the sublot shall be 75 ft long and 10 feet wide, with the centerline being placed on the right side of the trench.

A.	.627 x 75 = 47 ft	.595 x 10 = 6 feet left of centerline
B.	.458 x 75 = 34 ft	.137 x 10 = 1 foot left of centerline
C.	.510 x 75 = 38 ft	.656 x 10 = 7 feet left of centerline
D.	.324 x 75 = 24 ft	.284 x 10 = 3 feet left of centerline
E.	.135 x 75 = 10 ft	.903 x 10 = 9 feet left of centerline

WEST VIRGINIA DEPARTMENT OF TRANSPORTATION
DIVISION OF HIGHWAYS
MATERIALS CONTROL, SOILS AND TESTING DIVISION

MATERIALS PROCEDURE

GUIDE FOR QUALITY CONTROL OF COMPACTION

1. PURPOSE

1.1 This procedure sets forth minimum guidelines for the Contractor's Quality Control (QC) Plan for embankment, subgrade, pipe and random fill used as structure backfill material and aggregate base courses. It is intended that these requirements be used as a procedural guide in detailing the inspection, sampling, and testing necessary to maintain compliance with the specification requirements.

1.2 To establish procedural guidelines for approval and documentation of a Master QC Plan.

2. SCOPE

2.1 This procedure is applicable to all items requiring compaction control except asphalt pavements. This outlines the QC procedures for Compaction items and includes procedures for approving and using Master and/or Project Specific QC Plans. This procedure also aids in documentation and retention of QC Plans in ProjectWise.

3. DEFINITIONS

3.1 Moisture/Density Gauge (Gauge) – Any gauge that has been approved for use on WVDOH projects. A list of these gauges and their applicable uses is available of the WVDOH MCST Webpage.

3.1.1 Moisture/Density Gauge is to replace all instances of the term "Nuclear Gauge" in all WVDOH documents.

4. REFERENCED DOCUMENTS

4.1 MP 100.00.03 - Method Of Evaluation of Non-Standard or Non-Conforming Materials in Construction Via DMIR

4.2 MP 109.00.21 - Basis for Charges for Non-Submittal of Sampling & Testing Documentation by the Established Deadline

4.3 MP 207.07.20 - Nuclear Field Density - Moisture Test for Random Material Having Less Than 40% of +3/4 Inch Material

4.4 MP 700.00.24 - Nuclear Density Test by The Roller Pass Methods

4.5 MP 700.00.50 - Procedure for Monitoring the Contractor's Compaction Testing of Bituminous Concrete, Base Course, Embankment, Sub-Grade and Pipe and Structural Backfill

4.6 MP 712.21.26 - Procedure for Determining Random Location of Compaction Tests

5. GENERAL REQUIREMENTS

5.1 The Contractor shall provide and maintain a QC system that will provide assurance that all materials submitted to the Division for acceptance will conform to the contract requirements whether natural, manufactured or processed by the Contractor, or procured from suppliers. The QC Plan should clearly describe the methods by which the QC Program will be conducted. For example, the items to be controlled, tests to be performed, testing frequencies, sampling locations and techniques all should be included etc. Each item should be listed separately.

5.1.1 A detailed plan of action regarding disposition of non-specification material shall be included. Such a plan shall provide for immediate notification of the Division in the event of a non-conforming situation or instance.

5.2 Inspection and testing records shall be maintained, kept current, and made available for review by the Engineer throughout the life of the contract. All other documentation, such as date of inspections, tests performed, temperature measurements, and any accuracy, calibration, or re-calibration checks performed on production or testing equipment shall be recorded and kept.

5.3 The Contractor shall maintain standard calibrated equipment and qualified personnel in accordance with the contract and Specification requirements for the applicable material.

6. QUALITY CONTROL PLAN

6.1 The Contractor shall prepare a QC Plan detailing the type and frequency of inspection, sampling, and testing necessary to measure and control the compaction properties of materials and construction governed by the Specifications. As a minimum, the sampling and testing plan should detail sampling location, sampling techniques, and test frequency. QC sampling and testing performed by the Contractor may be utilized by the Division for acceptance.

6.1.1 A QC Plan shall be developed by the Contractor and submitted to the Engineer prior to the start of construction on every project. Acceptance of the QC Plan by the Engineer will be contingent upon its concurrence with these guidelines as listed in section 6.2 through 6.4.5.2.

6.1.2 As work progresses, an addendum(s) may be required to a QC Plan to keep the QC program current. Personnel may be required to show proof of certification for testing.

6.2 QC PLAN MINIMUM REQUIREMENTS

6.2.1 The QC Plan should be on Company Letterhead, be addressed to the District which it pertains, and include the items to be controlled. An example/template is provided in Attachment 1.

6.2.2 Provide the name of the Person who is responsible for the Company's QC program and will be liaison with the Division's personnel.

6.2.3 List all inspectors' names performing compaction tests on the project and their date becoming a Certified Soils & Aggregate Compaction Inspector as per WVDOH Specification Section 106 Control of Materials.

6.2.4 Compaction field tests will be performed according to MP 207.07.20, MP 700.00.24, and Standard Specification 716.32.3

6.2.5 Soft shale tests shall be conducted according to Section 716 of the Standard Specifications.

6.2.6 Specify in the plan the methods by which each item will be tested. Table A and Table B summarize the different materials, minimum frequencies, and the appropriate test procedure or method for controlling each material.

Table A- COMPACTION CONTROL OF AGGREGATE BASE COURSES

TEST PROCEDURE	LOT SIZE	NUMBER OF TEST	MATERIAL TYPE			
			PORTLAND CEMENT TREATED AGGREGATE BASE COURSE	CRUSHED AGGREGATE BASES AND SUBBASE COURSES	HOT-MIX HOT-LAID BITUMINOUS TREATED BASE COURSE	SOIL CEMENT BASE COURSE
MP 700.00.24	2000 FEET	1 PER SUBLOT 5 PER LOT	X	X	X	
MP 207.07.20	2000 FEET	1 PER SUBLOT 5 PER LOT				X

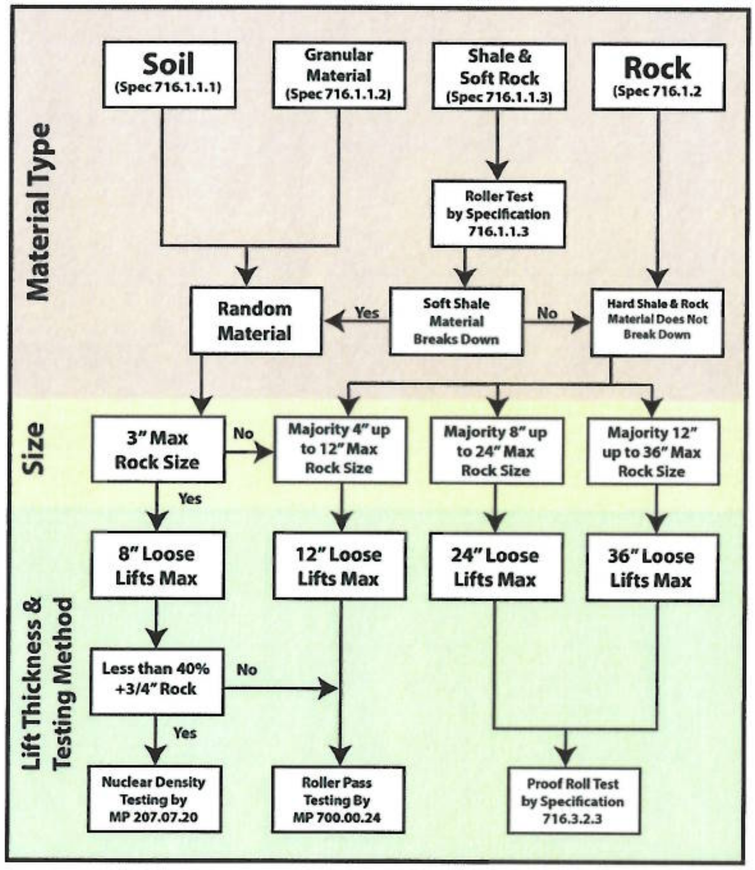
Table B - COMPACTION CONTROL OF EMBANKMENT BACKFILL AND SUBGRADE

TEST	LOT SIZE	NUMBER OF TESTS	MATERIAL WITH LESS THAN 40% RETAINED ON ¾" (19.0 mm) SIEVE	MATERIAL WITH 40% OR MORE RETAINED ON ¾" (19.0 mm) SIEVE AND CAN BE PLACED IN A 12" (300 mm) LOOSE LIFT OR LESS		MATERIAL THAT CAN BE PLACED IN A LOOSE LIFT GREATER THAN 12" (300 mm)		GRANULAR SUBGRADE	SELECT MATERIAL FOR BACKFILLING AND CLASS I AGGREGATE
				UNIFORM	NON-UNIFORM	ROCK	HARD SHALE		
MP 207.07.20	SEE STD. SPECS.	1 PER SUBLOT 5 PER LOT	X						
MP 700.00.24	SEE STD. SPECS.	1 PER SUBLOT, 5 PER LOT		X [1]	X [1], [2]			X	X
PROOF Rolling		1 REPORT PER LIFT				X	X		

1. If a hole for a direct transmission density reading cannot be readily made due to the coarse material, proof roll the lift.
2. If density readings vary above 105 percent or below 95 percent and the material appear to be non-uniform, proof roll the lift.

6.2.7 A flow chart for embankment material, Table C, shall serve as a guide for identifying material types, maximum rock size, lift thickness and compaction test method. This table shall be included in the QC Plan for making field decisions to ensure that each type of material is properly placed and compacted.

Table C – Guide for Quality Control of Embankment Material



6.2.8 The plan shall include a statement that all necessary testing equipment will be provided to perform the procedures outlined in MP 700.00.24, MP 207.07.20, and Specification 716.3.2. The plan shall list the make and model of equipment for proof rolling and its weight per Specification 716.3.2. The plan shall list the make, model and operating weight of the roller(s) to be used for the soft shale tests and per Specification 716.1.1.3.

6.2.9 List the type of gauge to be used. The calibration frequency must be acceptable to the Division. Gauges must be calibrated according to the manufacturer's requirements. This information shall be given to the Division upon their request.

6.2.10 If applicable, outline the procedure for performing a stability check on gauges that are not within the tolerance range for standard counts during the interval between calibrations. Standard counts derived during the stability check for stable gauges may be used in lieu of the manufacturer's standards. Gauges found to be unstable cannot be used until repaired and calibrated.

6.2.11 Include in the plan the lot and sublot sizes to be used for testing each type of installation. During construction, some flexibility in lot sizes may be made if the situation warrants in order to maintain a workable system. For example, two or more areas containing small quantities of embankment material might be combined into one lot at the Contractor's option and subject to the Division's approval.

6.2.12 Specify the maximum time period for completion of a lot of embankment material. As a guide, if the desired lot size cannot be obtained within seven calendar days, then the material placed up to that time would constitute the lot and the specified number of tests for a lot would still be performed.

6.2.13 Specify in the plan when quality control tests for base and subgrade will be performed. QC tests are to be performed after the material has been shaped and final rolling has been completed.

6.2.14 The Contractor is responsible for the accuracy of their individual testing and calculations.

6.2.15 List the forms and method of distribution for tests and measurements.

6.2.16 Compaction test results are reported on forms specified in MP 207.07.20 and MP 700.00.24. The forms are supplied by the Division and available on the MCS&T Webpage[1]. The completed form shall be submitted to the Division's project supervisor, District Materials Lab, and a copy shall be retained for the Contractor's records.

6.2.17 Indicate the length of time after tests and measurements are completed that documentation will be provided.

6.2.17.1 Test results and measurements shall be made available to project personnel for review on a daily basis. Formal submission of measurements should be made within 24 hours after the measurements are taken and test results within 24 hours after testing of a lot is completed.

6.2.17.2 Tests performed in a lot before final rolling is completed should be submitted to the Project Supervisor and retained in the project files. This includes test documents for failing lots and moisture checks.

6.2.18 List the compaction equipment giving the quantity, make, model, and weight or applied force at which each roller will be operated. If ballast will be added to a roller, indicate the type and quantity of ballast and the method for verifying the gross weight. Attach the manufacturer's specifications for compaction capabilities for each roller to the plan or state the procedure for verifying the compaction capabilities of each roller in cases where the manufacturer's specifications are not available. This equipment shall meet the requirements of Section 207.7.5 of the Specifications.

6.2.19 Indicate in the plan that a minimum of a 10-ton (9.07 Mg) roller will be used for testing in accordance with MP 700.00.24 for soil and granular material only.

6.2.20 Rollers used to breakdown soft shale shall be in accordance with Section 716.1.1.3 of the Specifications and shall have a minimum of 1.5 tons per linear foot of roller drum width.

---

[1] https://transportation.wv.gov/highways/mcst/Pages/tbox.aspx

6.2.21 Specify the method by which proof rolling will be conducted on embankment materials. The materials to be proof rolled are summarized in Table B in Section 6.2.6.

6.2.22 List the number of passes to be made and corrective measures if soft areas are detected. Documentation should include the type of material, number of passes, and corrective action if soft areas are detected.

6.2.23 For equipment used for proof rolling explain how the gross weight will be determined for any ballast added to the operating weight. For alternate proof rollers, attach to the QC Plan the calculations used to determine that the roller meets specifications. Also, attach the manufacturer's specifications for all proof rollers to the Plan. The following calculation is used to determine if an alternate proof roller meets specifications:

ENGLISH	Metric
$c = \frac{\sqrt{(ab\pi)}}{2}$	$c = \frac{\sqrt{(ab\pi)}}{50.8}$

Where:

a = weight (force) on a single tire = pounds (kg x .009807 = kN)
b = operating tire pressure = psi (kPa)
c = weight (force) per inch (mm) width of tire = pounds per inch (Nm)
The weight (force) per inch (mm) width of tire must be equal to or greater than 1315 pounds (9.067 kN/mm).

6.2.24 Outline the procedure for notifying the Division when the test section in MP 700.00.24 will be performed. The Division shall be notified a minimum of 24 hours in advance unless other arrangements acceptable to the Division can be made.

6.2.25 Laboratory testing for random material is not required unless the material has unusual characteristics or differs from the soil and rock data used to develop the design. Testing to develop density curves, specific gravities, organic content, etc. may be required.

6.2.26 A list of test procedures is contained in Section 716 of the WVDOH Standard Specifications as a guideline for required testing should the need arise for random material.

6.2.27 Design a plan of action for the disposition of non-specification material, such as material with excessive moisture, excessive organic content, etc. These materials shall be stockpiled away from the embankment or fill placement areas. The Project Supervisor should be immediately notified in the event a nonconformance situation is detected.

6.2.28 List the method(s) and frequencies per Table D by which lift thickness measurements will be taken. If surveying of compacted lifts is not utilized, then the maximum loose lifts per Table C shall be measured.

TABLE D - LIFT THICKNESS MEASUREMENTS

MATERIAL TYPE	NUMBER OF MEASUREMENTS
EMBANKMENT	MINIMUM OF 3 PER LIFT
SUBGRADE	MINIMUM OF ONE PER 1200 FEET PER WORKING WIDTH
PIPE BACKFILL	MINIMUM OF ONE PER SIDE PER LIFT
STRUCTURE BACKFILL	MINIMUM OF ONE PER LIFT

6.3 TYPES OF QC PLAN

6.3.1 QC Plans which are intended for use on more than one project shall be defined as Master QC Plans. Section 6.4 outlines the procedures for Master QC Plan submittal and approval.

6.3.2 QC Plans which are intended for use on a single project shall be defined as Project Specific QC Plans. Project Specific QC Plans shall contain a cover letter which includes the following: project name/description, CID#, Federal and/or State Project Number.

6.3.3 A contractor may submit a Master QC Plan for field operations instead of a Project Specific QC Plan.

6.3.4 Once any QC Plan is approved for a project, the key date shall be entered in AASHTOWare software by the appropriate District Materials personnel. The first date entered shall be the date the Project QC Plan letter is received. The second date shall be when the District approves the QC Plan for use on the project.

6.4 MASTER QUALITY CONTROL PLAN

6.4.1 The intent of Master QC Plans is to facilitate the approval process in a more uniform manner. A Master QC Plan can be submitted to the Division/District by the Contractor when their work in a given District is routinely repetitive for the year. The Master Quality Control Plan is applicable for only the calendar year for which it has been approved.

6.4.2 The Contractor shall submit the Master Compaction QC Plan yearly to each District in which they have work in. If the Contractor does not have work in a given District for the year then no Master QC Plan shall be submitted to that District.

6.4.3 The District will review the submitted Master QC Plan and assign a laboratory reference number upon approval for future referencing. The District will acknowledge approval of Master QC Plan to the Contractor by letter (see Attachment #2 for an example), which will include the laboratory reference number and a copy of the approved Master QC Plan attached. This will then be scanned and placed in ProjectWise under the appropriate District's Org for that Contractor.

6.4.4 Once a project has been awarded, if a contractor elects to use the approved Master Compaction QC Plan on that project, the Contractor shall submit a letter requesting to use the Master QC Plan for that project. This letter must be on the Contractor's letterhead, be addressed to the District Engineer/Manager or their designee, and contain the following information: project number, CID#, project name/ description, type of Quality Control Plan and the laboratory reference number for the Master QC Plan (See Attachment #3 for an example).

6.4.5 The District shall review the referenced Master QC Plan to ensure that it covers all items in the project. If the referenced Master QC Plan is found to be insufficient for some items on the project, the District shall request the Contractor to submit additional information for QC of those items as an addendum on a project specific basis. When the District is satisfied with the QC Plan for this project, a letter shall be sent to the Contractor acknowledging approval (see Attachment #4 for an example), with the following attached: the Contractor's project QC Plan request letter and the Master QCP approval letter. This shall then be placed in the project's incoming-mail mailbox in ProjectWise.

6.4.5.1 A Master QC Plan that has been approved for project use shall be acceptable for the duration of that project, even if that project continues into subsequent calendar years, unless otherwise directed by the District.

6.4.5.2 For the use of Division Personnel, the District approval letter for this project must state the ProjectWise link to the referenced Master QC Plan for that Contractor. (i.e., WVDOT ORGS > District Organization #> Materials > Year>Master QC Plans...)

7. CERTIFICATION & ACCEPTANCE SAMPLING AND TESTING

7.1 The Contractor shall certify that compaction testing and sampling is in conformance with the approved QC plan, referenced MP's and referenced Standard Specifications in a letter format on the company's letterhead. The certification shall summarize what materials were encountered and the compaction method/lift thickness utilized. The letter shall state whether any deviations from the requirements of the QC plan, MP's, and Standard Specifications exist, and why.

7.2 Acceptance sampling and testing is the responsibility of the Division. QC tests by the Contractor may be used for acceptance.

7.3 The Division shall sample and test for applicable items completely independent of the contractor at a frequency equal to but not limited to approximately ten (10) percent of the frequency for testing given in the approved Quality Control Plan. Witnessing the contractor's sampling and testing activities may also be a part of the acceptance procedure, but only to the extent that such tests are considered "in addition to" the ten (10) percent independent tests.

7.4 MP 700.00.50, MP 207.07.20, and Specification Section 716.3.2.3 outlines the procedures to be followed for acceptance of compaction testing.

8. ABSENT TESTING OF MATERIAL

8.1 If the Contractor fails to perform testing of the material in accordance with the Contractor's Division Approved Quality Control Plan, payment for the portion of the item represented by the absent test shall be withheld, pending the Engineer's decision whether or not to allow the material to remain in place. Testing includes both performing the test and submitting the results as per MP 109.00.21.

8.1.1 If the Engineer allows the material to remain in place, the Division shall not pay for the material represented by the absent test. However, the Division shall pay for the cost of the placement of the material, including labor and equipment. The invoice or material supplier cost (if applicable), determined at the time of shipment, shall be used to calculate the cost of material when evaluating the total cost of labor and equipment.

8.1.1.1 If there is no material cost, the deduction shall be assessed on the tonnage of material represented by the missing test via a District Materials Inspection Report (DMIR).

Michael Mance

Digitally signed by Michael Mance
Date: 2024.08.23 14:35:26 -04'00'

Michael A. Mance, P.E.
Interim Director
Materials Control, Soils & Testing Division

MP 717.04.21 Steward – Pavement Analysis & Evaluation Section
MM:A
ATTACHMENTS

ATTACHMENT 1 - EXAMPLE GUIDE FOR COMPACTION QUALITY CONTROL PLAN

The Acme Company
20 First St.
Somewhere, WV XXXXXXX

Mr/./Ms/Mrs. ________________________
WV Division of Highways
District ___ Engineer/Manager
__________, WV _________

RE: (YEAR) Master Compaction QC
Plan
DISTRICT: __________

Dear Mr./Ms/Mrs. ____________________

We are submitting our Compaction Quality Control Plan for field control, developed in accordance with sections 716 and 717 of the (year) WVDOH Standards and Specifications, _ (year) WVDOH Supplemental specifications, MP 700.0024, MP 207.07.20, MP 712.21.26 and MP 700.00.50.

The Quality Control Program is under the direction of ___________________ . They can be contacted by telephone number _____________, email ___________________ and/or in person.

1.) All testing will be performed by qualified personnel as per WVDOH Specification Section 106 Control of Materials. Proof of personnel certification shall be provided to WVDOH inspectors upon request.

2.) Specify the methods by which each item will be tested .(IE.. 207,307…etc). Table A and Table B (attached) summarizes the different materials, minimum frequencies, and the appropriate test procedure or method for controlling each material. A flow chart for embankment material, Table C (attached), is intended to serve as a guide for making field decisions to insure that each type of material is properly placed.

3.) Testing Equipment used will be as required in MP 700.00.24 and MP 207.07.20.

4.) Type of gauge to be used (IE.... Troxler 3430, etc). State that calibration information is available upon request by the Division/District.

5.) Outline the procedure for performing a stability check on nuclear gauges which are not within the tolerance range for standard counts during the interval between calibrations. Gauges found to be unstable cannot be used until repaired and calibrated.

6.) Include in the plan the lot and sublot sizes to be used for testing each type of installation.

7.) Specify the maximum time period for completion of a lot of embankment material.

8.) Specify in the plan when quality control tests for base and subgrade will be performed.

9.) List the forms and method of distribution for tests and measurements. (The forms are specified in MP 207.02.20 and MP 700.00.24.) State that test results will be made available to WVDOH personnel on a daily basis.

10.) List the compaction equipment giving the quantity, make, model, and weight or applied force at which each roller will be operated. If ballast will be added to a roller, indicate the type and quantity of ballast and the method for verifying the gross weight. Attach the manufacturer's specifications for compaction capabilities for each roller to the plan or state the procedure for verifying the compaction capabilities of each roller in cases where the manufacturer's specifications are not available.

11.) Indicate in the plan that a minimum of a 10 ton (9.07 Mg) roller will be used for testing as per 700.00.24.

12.) Indicate in the plan that when shale materials are encountered, the shale hardness test will be performed to determine if material is a soft shale as per 716.1.1.3 of the WVDOH Standards and Specifications.

13.) Specify the method by which proof rolling will be conducted on embankment materials. The materials to be proof rolled are summarized in Table B (attached).

14.) Laboratory testing for random material is not required unless the material has unusual characteristics or differs from the soil and rock data used to develop the design. Testing to develop density curves, specific gravities, organic content, etc. may be required. The Yearly Quality Control Plan should state that these additional tests must be performed by qualified Aggregate testing personnel as per as per WVDOH Specification Section 106 Control of Materials.

15.) Design a plan of action for the disposition of non-specification material.

16.) List the method(s) and frequencies by which the lift thickness measurements will be taken.

Very Truly Yours,

____
Title

ATTACHMENT 2

**** WVDOH LETTERHEAD ****

THE ACME COMPANY INC.
20 First St.
Somewhere, WV XXXXX

RE: Compaction Master QCP
Description: 20XX Year

Dear Mr./Ms/Mrs. ____________________,

Your Master Quality Control Plan(**M# - ######**) for Compaction has been reviewed and found to be acceptable for the following items:

- 207001-001 Unclassified Excavation		- 207002-001 Subgrade
- 211-001	- 307001 Items	- 604 items
- 212 Items	- 605 items	-etc....

As work progresses throughout the season an addendum(s) may be required to this QCP to keep the QC program current. **Please use M# - ###### when corresponding about this QC plan.** Please make sure that all appropriate personnel have a copy of this plan in their possession.

Very Truly Yours,

____
Title

ATTACHMENT 3

The ACME COMPANY
20 First St.
Somewhere, WV XXXXX

************ **EXAMPLE** **************

Mr./Ms/Mrs ______________________
WV Division of Highways
District ___ Engineer/Manager
__________, WV ___________

RE: Compaction Quality Control plan
for Field ---- Project

Fed. Project No ____________________________
State Project No. _________________________
Contract ID No. _________________________
Description ____________________________

Dear Mr./Ms/Mrs. ____________________,

We would like to use our approved Yearly Master Quality Control Plan, reference number _____________ for the project referenced above. All Compaction items on the referenced project are covered by the Master Quality Control Plan.

The QC Plan is under the direction of ___________________________, _______________(title), and will be the company's contact representative to the Department of Highways District Materials and Construction Departments. He/She can be contacted in person at the project, by telephone ______________________________ or at email account ________________.

Very Truly Yours,

____
Title

ATTACHMENT 4

**** WVDOH LETTERHEAD ****

THE ACME COMPANY INC.
20 First St.
Somewhere, WV XXXXX

RE: Compaction QC Plan
Project CID#: ########
Fed/State Project #: NHPP- ## - ####.##
Description: Falling Slide
County : XXXXXXX

Dear Mr./Ms/Mrs. ___________________,

Your request to use Master Quality Control Plan **(M# - ######)** for compaction on the project referenced above, has been reviewed and found to be acceptable for the following items on the referenced project:

- 207001-001 Unclassified Excavation   - 207002-001 Subgrade
- 307001 Items   - 604 items   - 212 Items   -etc.…

As work progresses throughout this project an addendum(s) may be required to this QCP to keep the QC program current. **Please use M###### when corresponding about this QC plan.** Please make sure that all appropriate personnel have a copy of this plan in their possession.

For Division/District
The Master Quality Control Plan can be reviewed in ProjectWise folder shown below:

WVDOTORG> D0# > year > MASTERQCPLANS > Contractors >Contractor Name > Name of Quality Control Plan

Very Truly Yours,

____
Title